PICTRETION – Mobile Application for tree Identification using Image processing with Mapping

ABSTRACT

Trees in the Philippines have essentially contributed to sustaining the need for wood and crucially the survivability of endangered species. The Department of Environment and Natural Resources (DENR), the Philippine office governing the environment, helps preserve native and production trees through strict supervision operated under their active laws and policies. A tree inventory form is provided by the office to register known trees around a forest land or community upon the request to cut down trees or tree preservation. Some of the form's contents include the identification of tree species with their geographic location. However, the current method of foresters relies on their knowledge and memory with no instruments or lacks the technology for the identification and direction of the trees. Using the mobile application PICTRETION, the manual process of identifying tree species and their location has been assisted through the use of an Android smartphone. The image of the tree leaf captured by the phone undergoes image processing through Python. The mobile app is capable of identifying only 5 species of trees namely: Acacia, Banaba, Mahogany, Molave, and Yemen. The application allows the user to send a tree identification request to the server wherein it will be processed, and return with information confirming its identified tree specimen. The confirmed request will be registered accordingly to its specific tree database and plotted graphically through Google map directing users to the tree location. The results of this study show that the system achieved 91% accuracy from the 100 samples conducted, certainly proving to have a good score. Thus, the system proves to be a reliable and beneficial utility for the identification and plotting of trees in the environment as a potential starting point for our government foresters and the development of this technology.

Keywords: DENR, image processing, database, google maps, OpenCV-Python

EXECUTIVE SUMMARY

PICTRETION is a mobile application of leaf identification in Android smartphones. It serves as an aid to foresters with their manual process of identifying tree species with geographic location. The image of the tree leaf captured by the smartphone will be sent through a private server computer and will be processed through image processing using Python. The mobile app is capable of only identifying 5 species of trees namely: Acacia, Banaba, Mahogany, Molave, and Yemene. Once identified by the application and confirmed by the user, the data is registered to each specific tree database and plotted in a personalized Google map. The Android application can run on devices that meet the required minimum version provided by the platform with GPS and Wi-Fi included.

The results of the study show that the proponents were able to display the identified trees with their location. The application was also able to save tree information through the network and records tree data to specific databases. The proponents were also able to successfully process the leaf image on varying leaf scenarios and weather conditions around the environment and classify its leaf identity processed using image processing and machine learning technique. The application had strengths in accuracy and reliability as well as the user-friendliness of the system, with a rating of good and fair for those objectives. The weakness of the system, in terms of responsiveness as the preprocessing of the image, consumes process time in the uploading and downloading of the data. In addition, the system only utilizes 25 samples of leaves per tree species as predefined data that somehow affects the accuracy identification negatively.

The conclusion of the study shows that capturing sample leaf images under raw background will not properly identify the exact tree species due to the misleading factors that causes misidentification. The Pic-A-Tree system was able to achieve an accuracy of 91% and reliability of 93.20%, and not only does the system may benefit the environment through the identification and plotting of the trees, but it can also be used as an initial technology to improve the methods applied by our government foresters.

Chapter 1 - THE PROBLEM AND ITS BACKGROUND

A. Introduction

According to (Oliva, 2007) trees are all around us. Even within our urban landscape trees can still be found. Beyond the cities, there are plantations of trees that supply us with wood. The wood gathered from these plantations is used for many different things. The amount of wood collected is tallied through tree inventory.

Based on (L.-C. Tan and P. B. Durst, 2018)The Philippines has the dubious distinction of being one of the first tropical countries to have its forest area severely reduced in large part by modern logging technologies and agricultural expansion. The idea of private tree farming was well received. Knowing the fact that the government could not protect forests even with well-armed guards. Private forest ownership was the hope to solve the problem of husking. The only problem was that no private farmer had the funds to start a very uncertain tree-growing business, which requires at least 8 years for final harvest and income.

According (L.-C. Tan and P. B. Durst, 2018) by 1972, the world bank was convinced by the feasibility of tree farming and allocated millions of dollars to support the program. It was called the "Tree plantation Lease Agreement". Because of the strong support from the local and international communities, many more farmers began to be interested in growing trees. By 1976, the initial 20 families grew to more than 300 families 1981 and the plantation area increased from 20 hectares to more than 4000 hectares.

According to (L.-C. Tan and P. B. Durst, 2018) tree farming is an accepted alternative to traditional agriculture in the provinces of Agusan del Norte, Agusan del Sur, Davao del Norte, Surigao del Norte, Surigao del Sur, Misamis Occidental, and Cotabato. In these provinces, there is now more than 100,000 ha of forest plantations. For the most part, this area is both the last bastion and frontier of forestry in the Philippines.

According to (Roweda, 2018) one chief forester of the Department of Environmental and Natural I has purplish flowers and its leaves are resistant to termites and Lycus beetles. Molave is often used in reforestation projects in the country. Ilang-ilang can grow to up to 40 meters tall and can be used for perfume extraction. The leaves of Banaba are widely used in the Philippines and other east Asian countries. Each flower has six white to purple petals and blooms only once a year. Banaba is also one of the 69 herbal plants promoted by the Department of Health (DOH). Acacia is the most widespread tree in the Philippines. It can be seen growing along roads and rice fields and is commonly used in plantations.

According to (M. Zaid, 2015)Previous projects and studies have shown that there are different ways to be able to identify the type of plant but end up using the leaf of plants as the basis for its species. These projects focused on delivering the type of plant to the user and were also able to show more information about the plant by analyzing the image of the leaf. A study shows the process in which it analyzes the leaf to identify its species. The process includes the application grayscale conversion towards the image then edge detection and dilation. After which, they select a part of the leaf and apply the "Inside gray-level co-occurrence matrix" to classify its tree specimen and determine its credibility.

Most research studies were focused more on plant families, but fewer studies focus on tree families due to its constraints in the height and the difficulty of obtaining fresh leaf samples. Since both tree and plant belonged to a similar family or branch in science and their leaf characteristics are almost interchangeable, the identification of tree species based on leaf structure may be possible soon. The fact that the idea impacts the environment and the government's management for tree preservation proves that the development can be meaningful.

B. Background of the Study

According to DENR Chief Forester, the responsibility of the office is to produce a tree inventory form if a request to cut down a tree is issued. The manual process of identifying the species of a tree is implemented since there is no instrument provided to the foresters when identifying tree species. Foresters must rely on their knowledge, memory, and landmarks to evoke the particular aspects of trees to identify forest reserves or forest areas, which sometimes causes inconsistencies when identifying the type of tree and its location. The actual method of identifying tree species perceived to be time-consuming and inconsistent especially on tree plantation and forest classification since some trees are confusing or have similar perspectives. Research studies also point out other aspects regarding the client's challenges on tree species identification in the validation of forest type based from (A. J. H. Ballado, 2017). The fact that there are few mobile species identification applications dedicated on smartphones, a need for an Android application using digital image processing and computer processing, must be introduced, or redeveloped. The idea is innovative since Android has adaptable and supportive technologies that can integrate certain functions and applications. Therefore, developing an interactive and responsive mobile automated application would hopefully solve the problem exerted on the conventional method.

In this case, the proponents aim to develop a tree classifying and locating Android application connected to a server-based system to assist the DENR in identifying the species and the location of trees that will be displayed on a tree mapping module. They also focused on the five common, naturally growing, and production trees in CALABARZON area in the Philippines, which are reserved and protected due to its endangerment issue or denudation problem within the Philippine Forest regions. The application will be Android compatible only to help connect and support with other platforms containing algorithms and features for image processing and tree identification. In this way, the system will make the current process more comprehensive, accurate, reliable in recognition, consistent, and response efficient.

C. General Objective

To develop a **PICTRETION-** Mobile Application for tree Identification using Image processing with Mapping

Specific Objective

1. To create an accurate system for the identification of tree species
    - Acacia, Banaba, Mahogany, Molave, and Yemen.
2. To develop a reliable system in leaf image identification
3. To design a user-friendly graphical user interface with a mapping module
4. To determine the consistency of tree species at varying distance
5. To develop a responsive system throughout the process of leaf identification

Constraints

1. The mobile application must be able to:
    - Identify the species of tree.
    - Provide the longitude and latitude of tree location.
2. The mobile application must operate under the laws and policies of the Department of Natural and Environmental Resources.
3. Google Maps A.P.I is written in JavaScript.
4. The application must meet the I.S.O. (International Organization of Standardization) 26000 standard requirements to provide practical guidance related to making social responsibilities operational and respecting cultural, societal, environmental, and legal differences and economic development conditions.

Metrics Description and Scale Measurement

Table 1 Metrics Description and Scale Measurements

Objectives	Definitions	Metrics	Description	Scale
Accuracy	Ability of the system to identify the tree species using image processing	Percentage average or ratio of correct recognition and number of samples	Excellent	Above 96.5%
			Very Good	92.43% - 96.49%
			Good	88.36% - 92.42%
			Fair	84.30% - 88.35%
			Poor	Below 84.29%
Reliability	Precision of correct specie based on identified tree	Average based on average deviation of tree parameters of all tree species	Excellent	Above 99.14%
			Very Good	96.23% - 99.13%
			Good	93.32% - 96.22%
			Fair	90.41% - 93.31%
			Poor	Below 90.4%
User Friendly	Ability of user to easily navigate through the graphical interface.	Measured through survey using Likert scale form.	Excellent	5
			Very Good	4-4.99
			Good	3-3.99
			Fair	2-2.99
			Poor	1—1.99
Consistency	Ability of application to determine the species of tree at varying distance	Measured through 30 samples of the same tree species.	Excellent	100cm
			Very Good	76-99.99cm
			Good	50-75.99cm
			Fair	26-49.99cm
			Poor	1-25.99cm
Responsiveness	The response time of a request to complete a system process	Average time of successful requests	Excellent	~1s-2.13s
			Very Good	2.14s-3.27s
			Good	3.28s-4.41s
			Fair	4.42s-5.55s
			Poor	Above 5.56s
	The rate of network failure to respond for requests	Percentage average of failed requests	Excellent	0%-0.92%
			Very Good	0.93%-1.85%
			Good	1.86%-2.78%
			Fair	2.79%-3.71%
			Poor	Above 3.72%

The table 1 shows and defines the metric that will be used to grade each objective. The first objective focuses on the accurate measurement of identifying tree specimens based on their leaf. The computations used to determine the accuracy of tree identification according to its leaf are shown below. Equation 1.1 is used to determine the maximum percentage scale, while the minimum value was computed using equation 1.2.

$$Ai = \left(\frac{total\ correct\ samples(t)}{number\ of\ samples\ tested\ (n)}\right) x\ 100$$

$$Accuracy = \left(\frac{TP+TN}{TP+TN+FP+FN}\right) x\ 100$$

Where variables for a minimum are True Positive (TP), True Negative (TN), False Negative (FN), and False Positive (FP) correlated to the results in the confusion matrix. The next objective defines the reliability of the application of identifying trees from image processing and average deviation is applied using each tree percentage parameter according to the ratio of the samples to determine the precision scale of the system on how well it identifies trees by species [13]. The equation below is applied to compute the average deviation for the reliability scale.

$$Average\ Deviation = \frac{\sum|X-\bar{X}|}{N}$$

Where the X variable refers to the scores of leaf feature vectors, the $\overline{X}$ is the mean of feature vectors and N is the number of scores. The next objective is regarding the user-friendliness of the application on how it is time-efficient in usage, presentable, usable, simple, robust, and responsive. It is assessed through a survey in Likert form that will be submitted to the client, while consistency is measured with 30 samples of leaf images with varying distances. Optimum results can be obtained from a shorter distance between the camera and the focused leaf. Lastly, responsiveness measures the response time of the application or system to process a request or the network request failure rate failing to respond to the user's request. Using Equation 1.1 located above, the parameter is measured by using the averaging method to average the time successful request and percentage mean of failed requests. SW

D. Significance of the Study

- **To Foresters**

This research will significantly help the people who protect and manage the forest. This will make the forester feel convenient and make every task easier. This will help them to speed up the process of identifying species of the tree. Furthermore, it will also become the storage of data that has been collected from the site.

- **To the DENR**

This will benefit the agency from gathering data from the forest they needed with ease. An advantage that will help them to control the forest, prevent damage, and losing trees. This will help them to improve management and to develop the environment**.**

- **To the Community**

This will aid the community to improve our environment. This will guide them in knowing the importance of trees. Acquire knowledge of the different types of trees.

- **To Future Researchers**

This is an opportunity for future researchers to have some ideas that are needed to improve the study of the same field. This will aid researchers in developing studies related to image processing. This study will give them broad insight into using a mobile application for data acquisition.

E. Scope and Delimitations

Scope

- Identify tree species through image processing of tree leaf.
- Provide the longitude and latitude of a specific tree.
- Generate a report consisting of the number of trees recorded in the system.
- Locate tree by specific tree I.D
- Provide a google map display with a tree log.
- Reference of species is provided through “Supervised Learning.”
- Focuses only on the forest located in region 4A (CALABARZON)

Delimitations

- The application is limited to android devices.
- The application will not work offline.
- A complete image of a whole leaf must be provided.
- The accuracy of the location is beyond the scope and depends on the GPS technology of the phone.
- Reliability of data transfer relies on internet connection.

Chapter 2 CONCEPTUAL FRAMEWORK

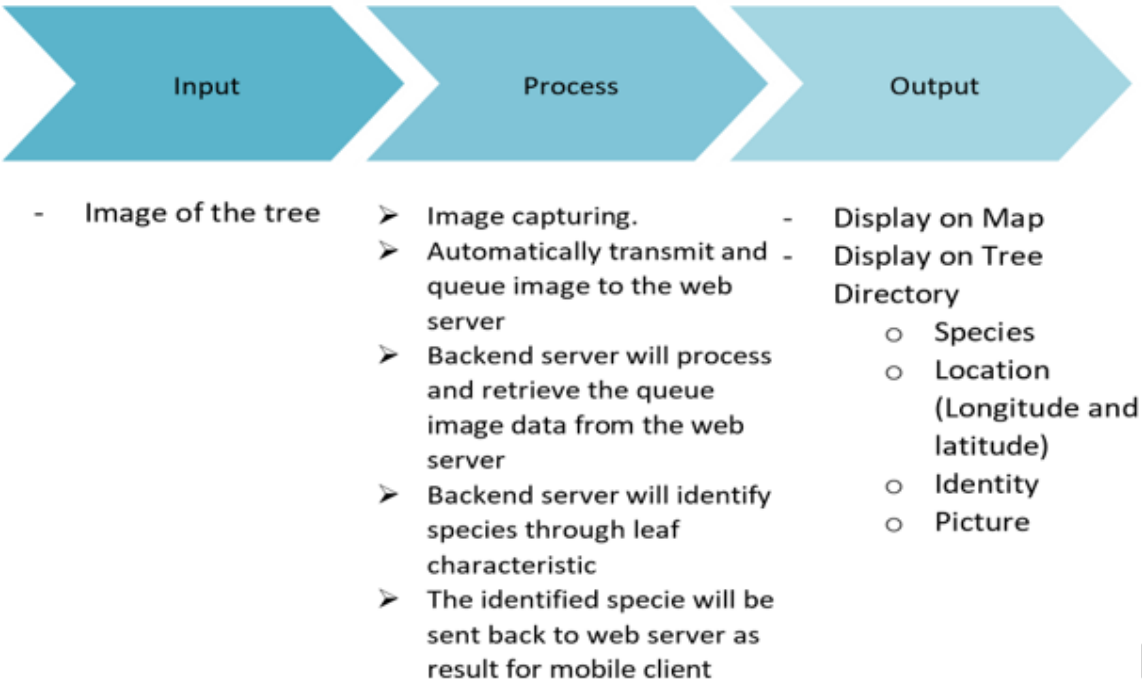


Figure 2.1 Conceptual framework

Figure 2.1 illustrates the input, process, and output of the proposed system. The system identifies tree species based on the photo taken leaf image and the location o where it resides. The exact location will be determined once the user captured the image of the tree leaf proximately close to the tree itself. After which, the confirmation of the taken photo will be automatically sent to the server for processing. After the server receives the queried image, leaf features will be extracted, then it will compare the extracted features with the basic feature from the training dataset. The result will transmit back to the mobile client that sent the request for leaf identification. As a byproduct, the data will be displayed as an output of the tree species with its geographic location.

Local Literature

Common and Native Trees of the Philippines

Based on (Celine, 2018) article discusses some essential and usually observed trees and can mostly be native trees inhabiting the country Philippines. It stated here that common Philippine inhabiting trees, like the Acacia, Golden Shower, Balayong, Fire Trees, and Caballeros, are foreign trees that were introduced and cultivated in the Philippines to increase the species survival due to their fast-growing characteristic. Though foreign, it is still a common tree usually found continuously and naturally growing in any urban and forest areas of the country.

Meanwhile, the common and native trees are first Katmon (Dillenia philippinensis)

Molave (Vitex parviflora), Bani (*Milletia pinnata*), Ilang-ilang (*Cananga odorata*), Banaba (*Lagerstroemia speciosa*), Malubalak (*Bombax ceiba*), Dap-dap (*Erythrina orientalis*), Salingbobog (*Crateva religiosa*), and Narra (*Pterocarpus indicus*). Katmon, first of all, is a native tree that is favorite by garden enthusiasts, suitable for furniture, and memorable for its fragrant and flavorful flower and average altitude. As for the Molave, it can be found anywhere in the Philippines, especially in the forests, though presently a dedicated native tree of some countries. It is famous by its strong wood features against parasites, such as termites, and considered as one of the hardest wood due to its strength, volume, and durability suitable for construction. Bani is a tree named from a town in Pangasinan, despite native to some other countries, has a large canopy and a cluster of fragrant flower, and can also produce other natural elements or resources, such as oil, soap, or lubricant. Ilang-ilang is a native tree and to some other countries and has the growth characteristic of reaching a height of 40 meters with a long-stalked petal that produces attractive fragrance. Banaba is a native tropical tree found in southern regions, and also in other subtropical countries. It has six purple petals and used as herbal plants by the DoH. Malubalak, though easily resembled with the other foreign and common trees found in other countries, are extremely fragrant from its red flowers and attractive to living creatures, including the humans. Dap-dap is a thorny and deciduous tree growing up to 27 meters tall, uniquely, found only in the Philippines, and grows large, numerous, and colorful petals that can be used for local medicine. Salingbobog has a mutual connection with the Cherry Blossom and is a moderately-sized deciduous tree that produces a secreting flower, fruit rich in vitamin C, and can also be treated for certain conditions. Finally, Narra, a native and commonly favored hardwood, used by furniture making industries, famous for its resistance to termites, and fragrant and attractive flower that secretes honey.

In this case study, the scope of tree species that will only be covered are the commonly observed and naturally occurring tree in the Philippines, which are the Acacia, Banaba, and Molave will be considered for the efforts of achieving the objectives and purpose of the study.

Acacia

The scientific name of Acacia is *Samanea saman*, translated in Spanish-Tagalog, or Akasya, natively translated in Tagalog-Ilocano. Acacia is a large umbraculiform tree growing to a height of 20 to 25 meters. The bark is rough and furrowed. Branches are widespread. Leaves are evenly bipinnate and hairy underneath. Pinnae are 8 to 12 and 15 centimeters long or less. Leaflets are 12 to 16 in the upper pinnae, 6 to 10 in the lower ones, decreasing in size downward, hairy beneath, with the mid-nerve diagonal, and oblong-rhomboid, 1.5 to 4 centimeters long. Flowers are pink, borne in dense, peduncled, axillary, solitary, fascicled heads. Fruits are pods, straight, somewhat fleshy, indehiscent, 15 to 20 centimeters long, 2 centimeters wide, with a pulpy sweet mesocarp.



Figure 2.1. Acacia tree

It is actually by spreads throughout the Philippine urban places like along roads, within the neighborhood, and trails of rice paddies. It is widely planted as a shade and can be found most in forest lands. Its constituents are abundant in alkaloids extracted from its bark, leaves, leaves, and stems, and starchy and sugary pods.

Molave

Vitex parviflora, commonly called Molave, is a tree that reaches a height of 8 to 15 meters, smooth or nearly so, with inflorescences that may be slightly hairy. Leaflets are three, stalked, ovate to lanceolate, 7 to 18 centimeters long, pointed at the tip, shining, and quite smooth. Flowers are blue, numerous, 6 to 8 millimeters long, hairy outside the corolla, borne on the terminal, paniculate and ample inflorescences, up to 20 centimeters in length. The fruit is rounded, 5 to 6 millimeters in diameter.



Figure 2.2. Molave tree

Standing at a low altitude, it is a native and common tree found in most forest section areas throughout the Philippine islands and provinces. It is known in the Philippines as one of the extremely hardwoods with its resistance to parasites and fungal attacks. This tree contributes well from its special properties on how it contains hydrocyanic acid and saponin in its fruit, bark, and root with antimicrobial and antifungal properties [19].

Banaba

Lagerstroemia speciosa, commonly called as Banaba, is a deciduous tropical flowering tree, 5 to 10 meters high, sometimes growing to a height of 20 meters. Bark is smooth, gray to cream-colored, and peels off in irregular flakes. Leaves are smooth, large, spatulate, oblong to elliptic-ovate, 4 to 8 centimeters in width, 12 to 25 centimeters in length, shedding its leaves the first months of the year. Flowers are 6-parted, purplish lilac or mauve-pink, rarely pink, 5 to 7.5 centimeters across, and borne in large, terminal panicles up to 40 centimeters in length. Petals are oblong-obovate or obovate, shortly clawed, and 3 to 3.5 centimeters long; the margins are undulate and hardly fimbriate. Fruit is a large nutlike capsule, obovoid or ellipsoid, and 2 to 3.5 centimeters long. Seed is pale brown, with a wing 12 to 18 millimeters long.



Figure 2.3. Banaba tree

Growing low and medium altitudes and can be commonly found inhabiting open forest lands in islands and provinces of the Philippines, provides an excellent urban tree due to the production of its beautifully attractive flowers. Aside from the secretion of compounds from its leaves, root, and fruit, it is a suitable tree for timber and other special-purpose wood.

The 2016 Statistics of Forest Population

According to (Natural, 2016) Philippine Forestry Statistics, the Philippines covers 14.19M hectares (ha) of alienable and disposable land and 15.81M ha of total forestlands within its 30M ha area of land.

Table 2.1. Status of Land Classification (in hectares)

Regions /Province		Total Area	Certified Alienable and Disposable Land	Total Forestland
Philippines		**30,000,000**	**14,194,675**	**15,805,325**
CAR		1,829,368	342,345	1,487,023
Region I	(Ilocos)	1,284,019	810,922	473,097
Region II	(Cagayan Valley)	2,687,517	972,822	1,714,695
Region III	(Central Luzon)	2,147,036	1,204,649	942,387

NCR	(Metro Manila)	63,600	48,232	15,368
Region IV-A	(Calabarzon)	1,622,861	1,051,948	570,913
Region IV-B	(Mimaropa)	**2,745,601**	**998,563**	**1,747,038**
Region V	(Bicol)	1,763,249	1,222,060	541,189
Region VI	(Western Visayas)	1,229,704	822,387	407,317
NIR		1,332,834	854,432	478,402
Region VII	(Central Visayas)	948,850	705,328	243,522
Region VIII	(Eastern Visayas)	2,143,169	1,024,955	1,118,214
Region IX	(Zamboanga Peninsula)	1,467,011	676,886	790,125
Region X	(Northern Mindanao)	1,714,803	817,669	897,134
Region XI	(Davao)	1,967,183	737,633	1,229,550
Region XII	(Soccskargen)	1,874,946	730,546	1,144,400
Region XIII	(Caraga)	1,884,697	544,897	1,339,800
ARMM		1,293,552	628,401	665,151

The Philippine forest areas and forestlands cover all of the regions of the Philippines from CAR, Region 1 to Region 13, and ARMM, and the MIMAROPA region has the largest area of forestlands with a forest population of 1.75M ha within the 2.75 ha of the land area followed by Region 2 and CAR. The focus of most tree researchers is forest reserves and plantations located in Region 4A, specifically in Los Banos and Mt. Makiling forestry, and Mimaropa as forest locations recommended by the Department of Environment and Natural Resources (DENR). The fact that they recommended the two forest area regions and locations because of the abundance of common, native, and naturally growing trees inhabiting within the prescribed area reserved and protected by the community, including the cultivation of tree plantations for forest resources. It also has a sense of convenience and a common ground for professionals, foresters, and researchers.

Table 2.2. Log Production (in cubic meter)

Region	Total	Falcata (Paraserianthes falcataria)	Mangium (Acacia mangium)	Yemane (Gmelina arborea)	Mahogany (Swietenia macrophylla)	Para Rubber (Hevea brasiliensis)	Gubas (Endospermum peltatum)	Bagras (Eucalyptus deglupta)	Ipil-Ipil (Leucaena Leucocephala)	Acacia (Samanea saman)	Others
Philippines	**790,098**	**480,163**	**101,650**	**80,266**	**70,546**	**8,371**	**5,628**	**4,929**	**1,645**	**1,691**	**35,209**

CAR	**885**	-	-	283	198	-	-	-	6	22	375
Region 1	**1,655**	-	3	371	618	-	-	1	7	226	428
Region 2	**740**	-	-	706	26	-	-	-	-	2	5
Region 3	**2,816**	-	-	932	923	-	-	40	26	291	604
Region 4A	**17,598**	-	-	1,695	11,824	-	-	-	-	30	4048
Region 4B	**20,440**	149	11,221	3,706	5,256	-	-	-	-	-	107
Region 5	**2,752**	143	3	380	1,586	-	-	-	-	218	422
Region 6	**13,704**	-	3,637	2,305	7,028	-	-	35	184	392	121
NIR	**1,839**	-	349	771	574	-	-	63	-	9	72
Region 7	**25,956**	-	398	3,677	20,126	-	-	-	65	1	1,689
Region 8	**49,707**	8	37,879	9,348	2,313	-	-	-	28	-	131
Region 9	**71,117**	8,080	21,220	27,918	7,307	477	-	1,680	12	3	4,417
Region 10	**47,349**	22,915	4,156	4,841	8,001	4,294	-	107	13	-	3,020
Region 11	**18,891**	13,340	235	2,420	348	944	-	356	430	-	816
Region 12	**22,817**	4,352	31	9,843	2,432	2,096	-	1,107	871	495	1,588
Region 13	**491,833**	431,173	22,518	11,070	1,980	558	5,628	1,537	-	-	17,367

Within Region 4A, according to (Celine, 2018) plantations of Acacia for log production or other forest resources are reasonably reduced since it is a common and natural growing tree found in the Philippines. The common and naturally growing trees in the Philippines, including Acacia, are Molave and Banaba. The Latter trees lately mentioned are dedicated species for the continuation of the efforts of this study. In contrast with Acacia, Mahogany has the highest population of 11,824 cubic meters of coverage within Region 4A. Reasons for not including the top common and native trees of the Philippines as plantations for log production is due to endangerments and vulnerability, which is now under the protection of the executive order with surveillance and preservation of government forest officials.

Foreign Literature

Image Processing

Image processing is a method to convert an image into digital form and perform some operations on it, according to (Garage, 2018) in order to get an enhanced image or to extract some useful information from it. It is a type of signal dispensation in which input is an image, like a video frame or photograph and output may be image or

characteristics associated with that image. Usually, an image processing system includes treating images as two-dimensional signals while applying already set signal processing methods to them.

It is among rapidly growing technologies today, with its applications in various aspects of a business. Image Processing forms the core research area within engineering and computer science disciplines too.

Steps of Image Processing

1. Importing the image with optical scanner or by digital photography.
2. Analyzing and manipulating the image which includes data compression and image enhancement and spotting patterns that are not visible to human eyes like satellite photographs.
3. Output is the last stage in which result can be an altered image or report that is based on image analysis.

Purpose of Image Processing

The purpose of image processing is divided into 5 groups. They are:

1. *Visualization* - Observe the objects that are not visible
2. *Image sharpening and restoration* - To create a better image
3. *Image retrieval* - Seek for the image of interest
4. *Measurement of pattern* – Measures various objects in an image
5. *Image Recognition* – Distinguish the objects in an image

Types of Image Processing

The two types of methods used for image processing are analog and digital image processing. Analog or visual techniques of image processing can be used for hard copies like printouts and photographs. Image analysts use various fundamentals of interpretation while using these visual techniques. The image processing is not just confined to an area that has to be studied but who with the knowledge of the analyst. Association is another

important tool in image processing through visual techniques. So analysts apply a combination of personal knowledge and collateral data to image processing.

Digital Processing techniques help in manipulation of the digital images by using computers. As raw data from imaging sensors from satellite platforms, it contains deficiencies. To get over such flaws and to get the originality of information, it has to undergo various phases of processing. The three general phases that all types of data have to undergo while using digital technique are pre-processing, enhancement and display, and information extraction.

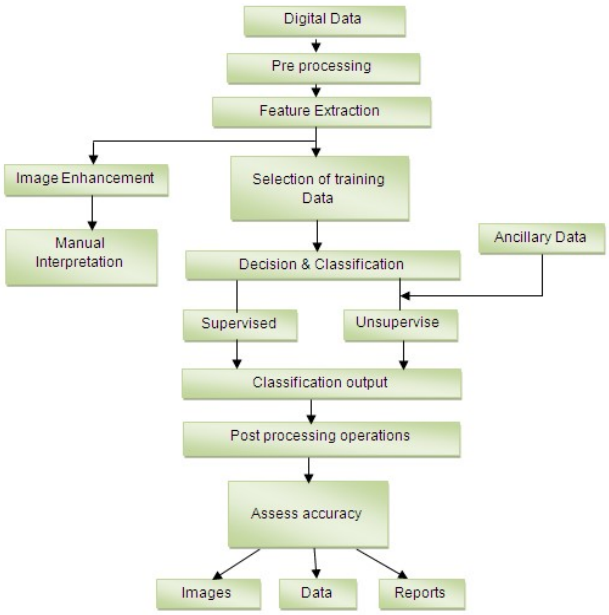


Figure 2.4. Complete system flow of image processing

Pre-processing

Based on (Kumar, 2018) a common name for operations with images at the lowest level of abstraction – both input and output are intensity images. An improvement of image data that suppresses unwanted distortions or enhances some image features important for further processing.

Feature Extraction [22]

Feature extraction is a type of dimensionality reduction that efficiently represents interesting parts of an image as a compact feature vector. This approach is useful when image sizes are large and a reduced feature representation are required to quickly complete tasks such as image matching and retrieval.

Feature detection, feature extraction, and matching are often combined to solve common computer vision problems such as object detection and recognition, content-based image retrieval, face detection and recognition, and texture classification. Deep learning models can also be used for automatic feature extraction algorithms. Other common feature extraction techniques include:

- Histogram of oriented gradients (HOG)
- Speeded-up robust features (SURF)
- Local binary patterns (LBP)
- Haar wavelets
- Color histograms

Classification

Image classification is the process of assigning land cover classes to pixels according to (Image Classification Techniques in Remote Sensing, 2018). In general, these are three main image classification techniques in remote sensing: unsupervised image classification supervised image classification, and object-based image analysis. Unsupervised and supervised image classification techniques are the two most common approaches. While the object-based classification has lately been used for high-resolution data, it will not be covered in the study since the focus is pixel-based and object-based are more appropriate in spatial resolution imagery,

Unsupervised Classification

- In unsupervised classification, it first groups pixels into “clusters” based on their properties. To create “clusters”, analysts use image clustering algorithms such as K-means and ISODATA. The unsupervised classification steps involve generating clusters and assign clusters repetitively.

Supervised Classification

- In supervised classification, one selects representative samples for each land cover class. The software then uses these "training sites" and applies them to the entire image. It uses the spectral signature defined in the training set. For example, it determines each class on what it resembles most in the training set. The supervised classification steps involve selecting training areas, generate a signature file, and classify until the next procedure is initiated.

Ideas of Assessing the User-friendliness of a Mobile Application

Based on (SmallBizDaily, 2017) Increasing as fast as mobile devices are good practices intended for the design of a mobile experience. When it comes to a vote, user experience wins any day. Creating a mobile app with a great user experience can be quite challenging, and the mobile app development continues to level up to that challenge becoming complex and assessment difficult. To achieve a good mobile experience, five ideas must be taken into consideration to better understand the expectation from a mobile developer's point of view.

Time Efficient in Usage and Ease of Navigation

Most of the time spent using the phone is during those times when one is involved with something else. The mobile app developer should remember that most people want to get done quickly whether they are glancing through their phones or they want to get something done.

Presentation

It is always a good practice to give space and a careful balance on the workplace to reduce load information in the screen. Questions one has to ask yourself during this development are necessary as they will assist in a clean and effective presentation of the app. Can they see the font? Can they read from the app effectively while in a hurry? Can they see the interface while under the sunlight? Are the buttons and links broad enough to click easily?

Ease of Learning and Usability

The app being developed has to be easy to use, and people don't need a manual to use it. The application needs to be simple and convey its capabilities. One can get a user interested within a short period it might be as brief as the time for a TV commercial. If potential new users can't download the app and figure out their way around the app in that period, one loses them.

Simplicity and Robustness of Design

Many books in libraries educate about good design. No matter what the definition gives to a good design, a good design is visible, transparent, and flawless.

Responsiveness

This idea plays a vital role in app development. No one will use the app if it is slow or heavy. No one will use an app if it does everything on one screen. Users tend to be impatient when they can't see the app loading or showing a movement as it leads them somewhere else. Speed is a feature that interests users, if you can't make it fast, make a progress bar to show time left until completion. Making an app with a great mobile experience is mostly an art than it is science. The process requires having the view of the users, doesn't matter if they are using their phones or a desktop, one needs to create an app that understands the balance.

Based on (Sinaian, n.d.)To assess this idea, a Likert form survey can be conducted with users regarding their experience and usage of the system to gain user's feedback.

Response Time in Mobile Applications

Agreeing to (Isakson, n.d.) on one of the very causes of downfalls in modern mobile applications is its actual response time. This component is essential for the optimum experience of the user accessing the data needed immediately. Mobile users these days are sharp and expect results in an instant. With mobile data response times communicating with fully operational or busy servers, quick response time is difficult to accomplish.

According to New Relic, there is an average of 639,000 active users per minute daily and within those users, an average response time of 3.28 seconds is accomplished.

With one tap of a screen, users will be able to communicate to the application and the smooth transition process are effortless. By tackling inside the system, every request digitally sent by the user will undergo a process in the background where it will encounter the backend service provider, searching for the required data and files in the database, and backend service returns the request with the data fetched operating within the server. For optimum performance, the application must execute the entire process within the 1-second timeframe.

Furthermore, the article openly disclosed that within the applications busiest minute from receiving a max number of web requests of 835K for an app with 35,600 active users at a time, an average requests rate of 1.86% will experience networks request failure and 1.7% will experience server error.

Local Studies

Plant Identification by Image Processing of Leaf Veins

Plant identification has always been a burden and laborious process to skilled botanists and biologists or professionals that specialize in plants, according to (L. Zhang, 2016) plant pathologists, plant breeders, and doctors that specialize in medical plants. With the use of the proposed timely device, the process of identifying and examining the basic information of plants will be automatic, helps reduce the effort of the plant sampling process to the laboratory, and improves from its limitation on record capacity.

The objective of the study is to integrate image processing algorithms to identify plants utilizing their leaf veins structure. Using Python language and raspberry pi processor, the prototype would be able to identify the plant's name and description in real-time and will be presented on the LCD screen. The captured or edited data will be updated into the database and thereby be analyzed and collected on the database through SIFT (Scale-Invariant Feature Transform) and SVM (Support Vector Machine) algorithms. SIFT analysis extracts and compares new features from features collected from other images in the database. Using Euclidean distance formula,

$$Euclidean\ Distance = \sqrt{(a_2 - a_1)^2 + (b_2 - b_1)^2 + \cdots + (n_2 - n_1)^2}$$

the matching candidate based on its feature vectors will be determined. While the SVM classifier serves as the voting system function to classify the leaf name. It votes out the different features compared from the data inside the SIFT database and returning the best match from the image classes for the class category of the leaf. The device's accuracy (ACC) and error rate (ERR) can be computed respectively using the formula below.

$$Accuracy = \left(\frac{TP+FN}{TP+TN+FN+FP}\right) x\ 100$$

$$Error = \left(\frac{FP+FN}{TP+TN+FN+FP}\right) x\ 100$$

It shows that the device accuracy for correctly identifying plant leaves is 84.29% while for incorrectly identifying plant leaves are 15.71%.

Forest Mapping and Classification of Forest Type using LiDAR Data and Tree Specie Identification through Image Processing Based on Leaf Extraction Algorithms

Based on (A. J. H. Ballado, 2017)Light Detection and Ranging (LiDAR) data, data originated from the Phil-LiDAR 2 Program, are utilized to derive the raw data and determine the Canopy Height Model (CHM), Canopy Cover Model (CCM), Digital Terrain Model (DTM), Diameter Breast Height (DBH), Above Ground Biomass (AGB), and Carbon Stock (CS) estimation for the construction process of forest mapping. Before the idea emerged, forest mapping manually has been a tedious task and requires strong memorization especially to forest rangers or dwellers that wishes to discover and identify a tree species and classify forest areas for forest type. Also, identifying numerous trees in a specific location within the area can be a challenging and skillful task due to the number of existing species and identifying characteristics of some tree species based on their leaves. With the use of LiDAR to plot a square meter of forest map and the use of Neural Network from MATLAB to perform image processing for tree identification, the two primary objectives of the research: mapping and classifying forest type, and identifying tree species in the province Batangas, Philippines.

The prototype contributes fairly to the development of forest mapping and monitoring through the use of high-resolution and advanced techniques for tree recognition based on its leaf. The study produces results based on its sequential data process workflow:

1) Organizing Datasets

- selected target area from LiDAR blocks under the datasets 2) Generation of CHM, CCM, DTM Layers
- Canopy Height Model (CHM) describes the height of the forest vegetation above ground, Canopy Cover Model (CCM) measures the percentage of the ground covered by the perimeter of plants, and Digital Terrain Model (DTM) describes the topographic surface of the Earth.

3) Generation of LiDAR perimeter bands

- shows the parameters generated by LiDAR as input features. 4) Classification
- Tree classification algorithm to classify forest types 5) DBH, AGB, and CS Estimation
- indirect estimations from LiDAR data to enhance the details for forest classification and mapping

6) Hardware and Software Development

- shows the leaf recognition process from the leaf to be digitally acquired by the scanner and save to the CPU, where the algorithm in MATLAB will process and extract the leaf's features such as leaf veins, areoles, edge detection, and area, and will be displayed in the LCD.

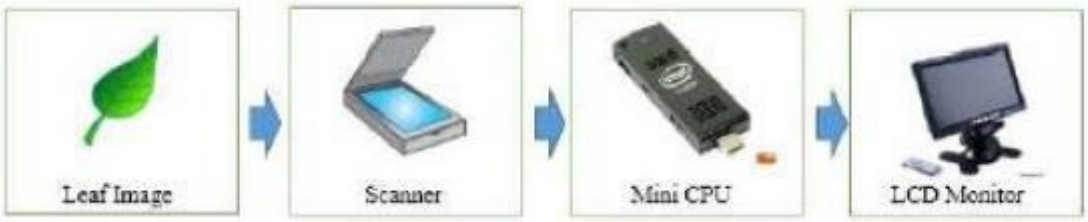


Figure 2.5. Tree species identification process diagram

7) Quality Check

- checks the output according to its descriptions 8) Map Layout
- displaying the final map layout and identified tree species in that designated area

Using T-test analysis, the actual perimeter is compared with the perimeter measure in the prototype's software tool to measure the prototype's accuracy.

Conducting five trials for each 10 out of 80 samples, the accuracy resulted in 91.25% with a minimum and maximum precision of the prototype to be 90.40% and 99.14% respectively.

$$Ai = \left(\frac{t}{n}\right) x\ 100$$

Where t – total correct samples
n – number of samples tested

Hence, the result is

$$Ai = \frac{73}{80} \times 100\ = 91.25\%$$

Mapping Mangrove Forest from LiDAR Data Using Object-Based Image Analysis and Support Vector Machine: The Case of Calatagan, Batangas

Based on (Jr, 2015)the production of forest maps for the conservation and environmental assessment of mangrove forests requires a lot of effort. Such required effort can be a cause of difficulty from the image capturing and detail analysis, time, and accuracy. To aid the efficiency and performance process of forest mapping, the study produces a high-

resolution scale map of selected areas in the Philippines. The study focused on the mangrove forest terrains in Calatagan, Batangas.

LiDAR (Light Detection and Ranging) data are used as image acquisition as input to process for image classification but before must undergo multiresolution segmentation enabled by the pre-processing stage. The best-segmented features to be observed CHM (canopy height model), average intensity, number of returns arithmetic. Using the SVM (Support Vector Machine) algorithm, the land and sea terrains can now be easily distinguished and classed of non-ground objects with ground objects. 14 features were selected for SVM training where eight features are from LiDAR-derived data and six features are from geometry features.

Applying the accuracy assessment based on the Test and Training Area (TTA), the accuracy obtained was above 0.87 and the overall accuracy produced from the fine-scale mangrove map at 0.91.

Philippine Forest and Wildlife Law Enforcement

According to (Oliva, 2007) the Philippines is the top mega country of a major contributor to the country's economy. During the 60's and throughout the '90s, their primary contribution was agriculture, mining, fisheries, and logging. Although the annual gross value earned from logging within was P15 billion from logging, it dramatically plummeted from 12.5% to 2.3%, and the Philippine forest was reduced from 30 million ha to 7.2 million ha of forest remains today.

As a reaction to the forest issue, the DENR (Department of Environmental and Natural Resources) is authorized and assigned to manage the Philippine forest areas, especially with the observed rising issue today. One of their actions is to identify and establish ground rules with the two major classifications of forest lands: protection forests (set-asides) and production forests. Protection forests are reserved and conserved forests for species of trees and other wildlife migrating within the area, while production forests are lands for tree production or plantation and rehabilitation for utilizing natural resources such as lumber and leaf processing. With help of other agencies, DILG and

LGU, and full implementation of the Philippine government laws such as the PD-705, the DENR forest management program and projects were made possible.

Foreign Studies

Authentication of Herbal Medicinal Leaf Image Processing using Raspberry Pi Processor

According to (Gopal, 2017) Herbal plants play a major role in the field of medicine. Plants have unique characteristics and qualities that serve for medical use but can have a similar outlook. Therefore, specialists and researchers examine the plants to better know their species and their characteristics for medical use. With the use of computer technology, image processing will be utilized to thoroughly authenticate the classification of a leaf from a plant and analyzed based on its features. A computer-based method will be applied to analyze the leaf biometric feature for the morphological feature analysis and artificial neural network-based classifier and Raspberry pi technology is the device that will serve as CPU for the pre-processing, feature extraction, and classification of sample leaf image.

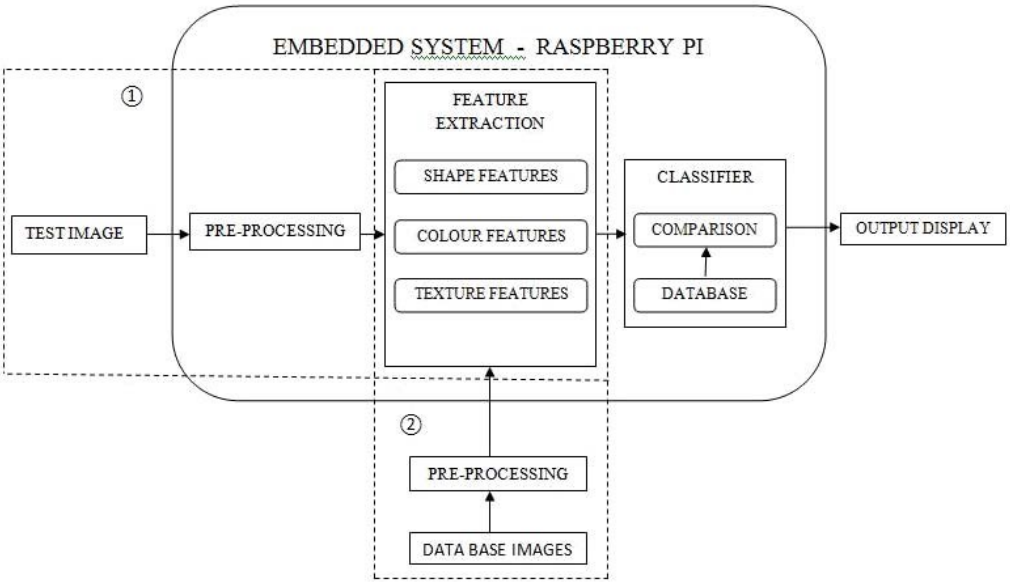


Figure 2.6. Block diagram of medicinal plant classification system

1) Preprocessing - the technique undergoes the methods enhancement and filtration for better identification, feature extraction, and leaf classification. The following methods that were used are listed below:
   - Greyscale conversion (1)
   - Edge Detection (2)
   - Dilation (3)
   - Fill image (4)
   - Selecting a part of leaf (5)
   - Finding gray-level co-occurrence matrix (6)

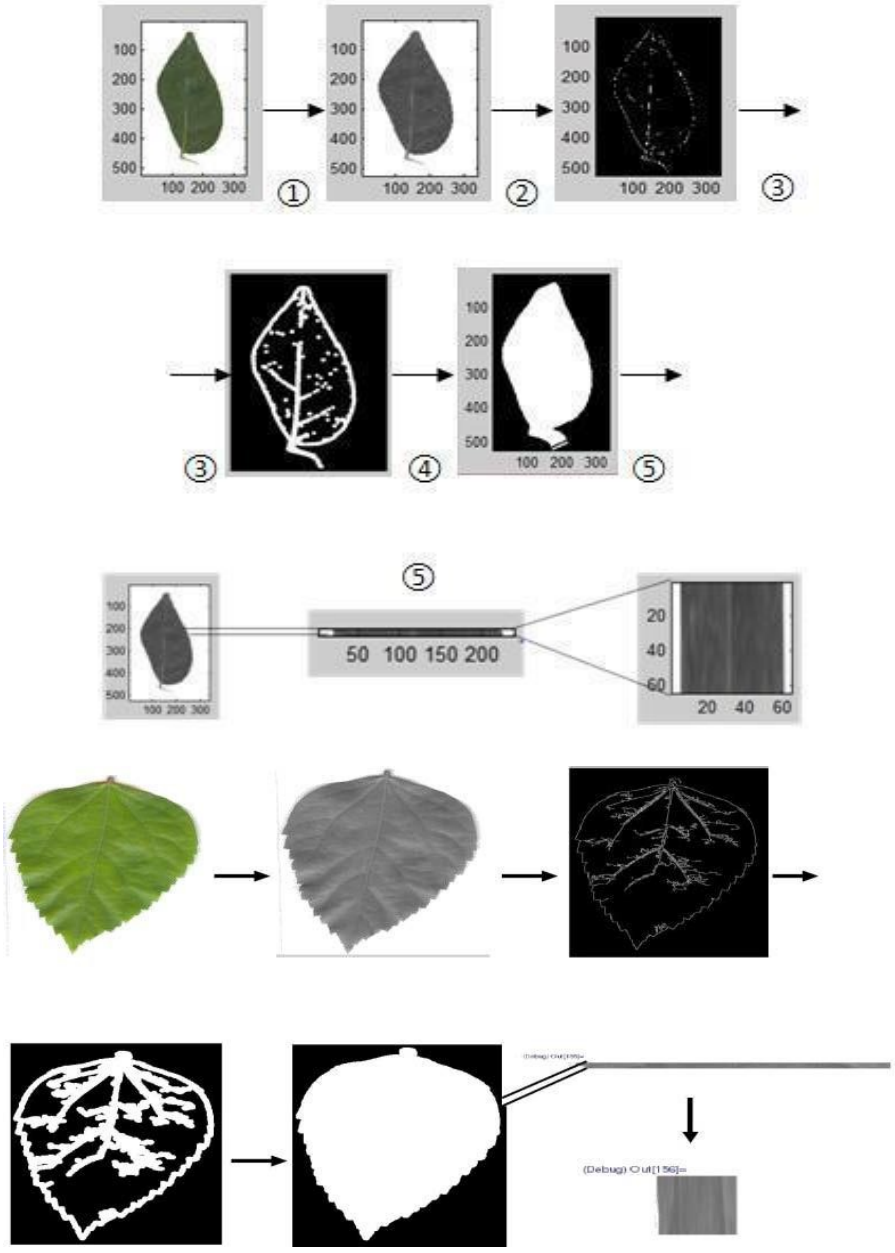


Figure 2.7. Pre-processing technique in Raspberry Pi

1) Feature extraction – from this pre-processed image, the feature extraction focuses on the parameters: aspect ratio, entropy skewness, edge detection, shape, texture, uniformity, inverse element difference moment, maximum probability, element difference moment, and vein feature.

2) Leaf classification – the herbal plant is classified by comparing the queried leaf image with the feature extracted leaf image from the database for training. The best match is based on the highest accuracy in the classification training process. PNN (Probabilistic Neural Network, k-NN, SVM (Support Vector Machine), and Fuzzy Logic are used as leaf classifiers.

The output will be displayed on the screen monitor connected to the device. Throughout the image processing, the monitor will present the extracted feature and the information of the leaf classification of the medical plant. The study did not conduct an accuracy test from the sample extracted to determine how well the system classifies plant leaves, but there stated that extracting of said feature will produce close to a 95-100% accuracy result.

Mobile Application for Quick Classification of Plant Leaf Based on Color and Shape

Based on (M. Zaid, 2015) the study presents a development of an application that uses mobile phone technology and image processing technique, it undergoes a classification algorithm to acquire the color scheme and feature of a leaf, thereby identifying the plant species. Raw images or photographs of a plant will be acquired using the camera provided by a smartphone. The method proposed uses preprocessing to obtain the binary contour of the leaf, morphological feature extraction, to obtain the feature vectors of the leaf, and finally, classification of the leaf based on the computer feature vectors. At first, the system must perform a training and testing phase to easily classify between known and unknown species, where the known species will be saved in the database. A system organization is presented to help readers comprehend the process definitions and the entire system.

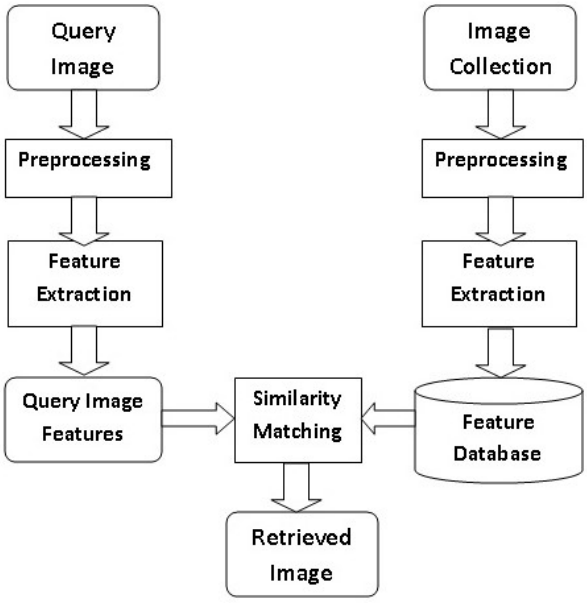


Figure 2.8. System diagram of plant classification application

Query image acquisition

Using the high-quality camera of the mobile phone, the plant's leaf will be taken, which will now serve as the leaf of the unknown species and input to be processed within the system. To avoid discrepancies, the image should be acquired at a degree of reasonable distance, decent lighting, and surfacing with high contrast background.

Preprocessing

The preprocessing stage obtains the contour feature of the original image. It locates the dominant color to be later used in the color matching phase. In the meantime, the image is converted into a bitmap image then converted into a binary image, i.e. black and white image, using Otsu's method and highlight the leaf's edges using the Sobel operator.



Figure 2.9. Sample leaf before and after preprocessing

Morphological feature extraction

After obtaining the control image, the system can now better detect the attributes or morphological features of the leaf. The study focused on the following leaf feature parameters in the algorithm: aspect ratio, rectangularity, form factor, sphericity, and eccentricity.

Training and matching

Five samples of each class were acquired for training the system, while three processes of matching the new image with the stored classes. The 3 matching process is the shape matching, where it is obtained from the binary contoured image performed in the preprocessing, the color matching that was dominant color obtained from the original image, and the morphological feature matching that involves the computation of the feature vector from its extracted features using the Euclidean distance.

Using the Java programming language, the algorithm was implemented in the Eclipse platform and will be further tested on the mobile phone. As a result, the system presented a fair classification accuracy limited by the five given sample classes. Improving its accuracy would require an extensive amount of sample classes but will drain the memory storage and performance of a smartphone. Therefore, the study recommends improving the system in the cloud-based environment where storage and classification are done through a cloud application.

Computer Assisted Plant Identification System for Android

According to (Withanage, 2015) Scale-Invariant Feature Transform (SIFT) features were described in the study that produces more accurate results than the conventional feature extraction. This descriptor algorithm is translated into the Bag of Words (BoW) model that helps segregate and classify leaf features accurately in the Support Vector Machine (SVM) classifier. With these methods, the system was able to produce an accuracy above 95% and proven to be the best against different plant or tree identification systems approaches.

The system proposed is an Android application that communicates to a server and gives users the ability to identify plant species using photographs taken of plant leaves using the smartphone. The study explained how the system works in its proposed architecture.

The mobile smartphone served as the Android client it will send leaf photographs as input and receive information about the leaf's identified species as output. The photo is uploaded to the server where the specific server process will perform to identify leaf species: the first is acquiring photographs uploaded by the Android client and generate feature vector and the second training of the SVM and generate the classification feature vectors and save it in the server file system. The diagram below illustrates the system as a whole.

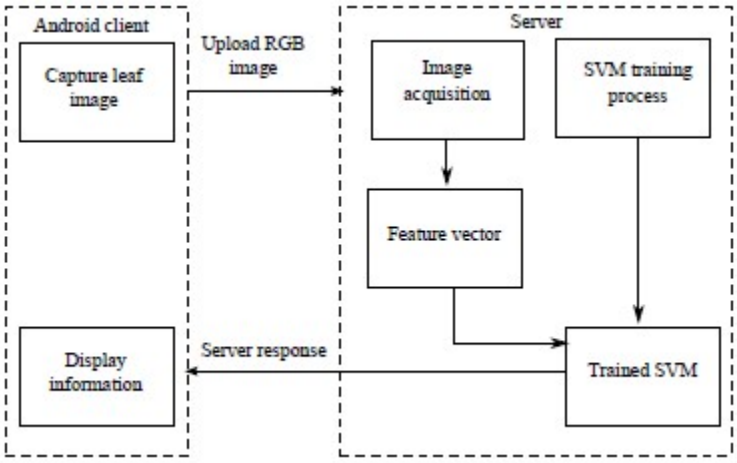


Figure 2.10. The overall architecture of the system

In extracting the features of the leaf, there are major steps to consider. First, SIFT descriptors combined with the BoW model provide training and testing datasets extracted from leaf images. As for the BoW model, it is an unsupervised learning method used as a dimensionality reduction method of the data space. Secondly, the BoW cluster all features in the data space (x-y coordinate), then BoW generate histograms that arranges training and testing datasets from the bag of vocabulary and bins of histogram containing the SIFT features, and finally, classify feature vectors from the histograms to the SVM, which will be saved as a classifier in the server file system. The diagram below shows the clustering and histogram creating sequential processes.

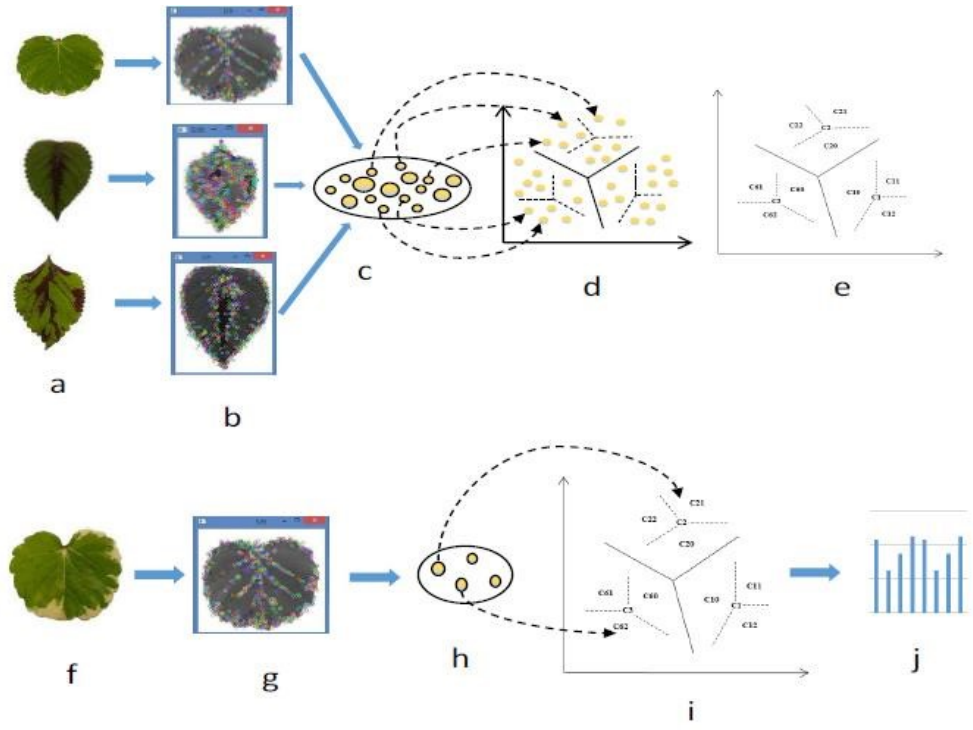


Figure 2.11. Generating classification feature vectors

Where:

a - Sample leaf images of train data set b - Obtained SIFT feature points c - Computed feature descriptors of key points d - Cluster feature descriptors e - Create and train the Bag of Features, build the vocabulary f - Take sample image from train data set g - Extract SIFT key points h - Compute SIFT descriptors for each key point i - Match feature descriptors with the vocabulary j - Compute the histogram

Before extracting SIFT feature points, the input images need to undergo preprocessing that converts the raw image into a grayscale image. Afterward, the key point can now be extracted and generate the descriptors from the grayscale image to produce SIFT features. The obtained SIFT feature descriptors will now be clustered in the data space using the k-means clustering method. The clustered SIFT features will have collected in respective

bins of the histogram, where they will be used as features to generate the classifier trained by the SVM. The diagram below illustrated the flow of the classifier method.

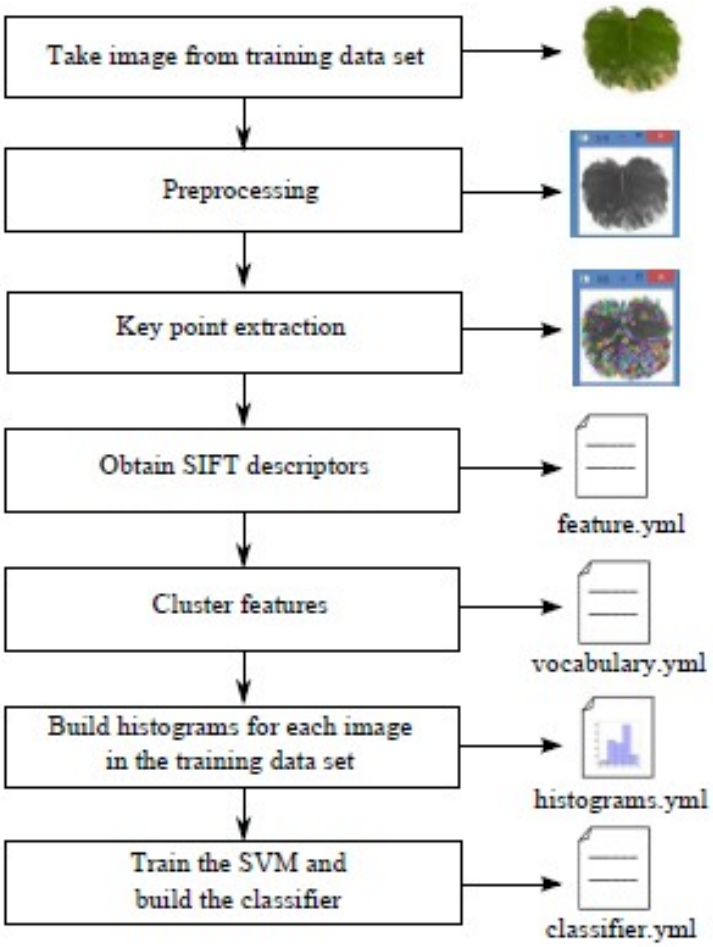


Figure 2.12. Method of creating the classifier

The Android application will only serve as the client whereas the leaf recognition algorithm will be running on the server. Information interchange and communication occurs over the internet using SOAP-based web service, which is shown below.

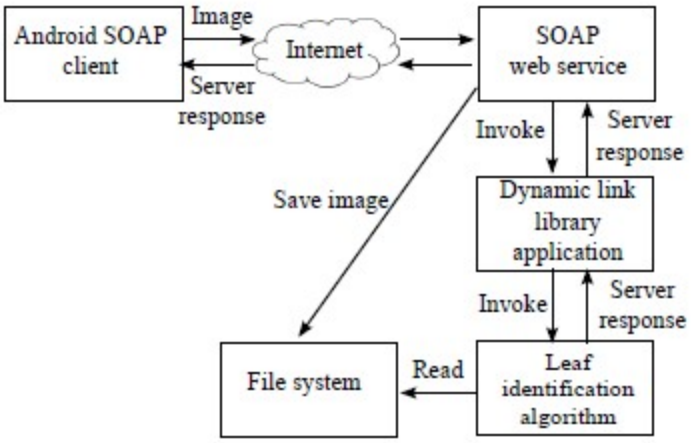


Figure 2.13. Client-server interaction

Through experimentation, the datasets are divided and organized depending on the purpose of the test. The experiment conducted 20 species of plants where each of the species contains 700. The 500 sample images were served as training and the 200 were served as accuracy testing. The accuracy is computed using the ratio of the correct recognition and the number of test samples. In this study, the accuracy obtained was 96.5%, which was a promising result compared with other feature extraction and classification approaches from other research studies. In a conclusion, the development of an Android application and the SIFT feature truly is an appropriate feature since it considers the feature extracted constant by in any scale and robust in object detection, which made the training process efficient and simple to implement.

Pl@ntNet Mobile Application

Based on (H. Goëau, 2014) Pl@ntNet is an image sharing and retrieval application for the identification of plants, launched on iPhone in 2013 and on Android devices in 2014. The application is flexible in the fact that it can perform with several parts of the plant or tree features: flowers, leaves, fruits, and barks. Data is collected to the database all credited to the collaborative workflow of the members of the social network specialized on the plants.

Compared with the iPhone application, LeafSnap, Pl@ntNet is now diversely compatible with other platforms like Android, and it is considered to be a promising solution of a

mobile application that helps setting-up a massive ecological monitoring system. Complementary with the worldwide data integration efforts and other information-sharing websites like the *Encyclopedia of Life*, the demonstrated app is built on an end-to-end collaborative information system allowing information within the application to be continuously enriched and revisited. The system application relies on the insight of professionals or tree or plant experts within the social network to solve the data validation issues and produce accurate knowledge so amateurs and experts can collaboratively validate data based on their observation and interactions together according to their level of understanding. They also play complementary roles for the real-world ecological surveillance workflow.

While in the 2014 version, the application is now compatible with the Android platforms, 3 times more data, exploiting metadata and visual content in the identification process, and new multi-image and multi-feature merging strategy with separate indexes for each visual feature, and integrating cross-languages functions. The application is also affiliated with ImageCLEF2013 plant or tree task provided by its abundant data of unknown and training set of plant or tree with leaf images.

Individual leaf identification from horticultural crop images based on the leaf skeleton

According to (L. Zhang, 2016) the study proposed a method of identifying the leaf skeleton and leaf distribution from horticultural crop images to eliminate the problem of complex scenarios due to varieties and obstructive instances, such as messy distributions, fuzzy edges, overlapping, and weeds from an image capturing and processing. It proved that the skeleton can be used to estimate the locations and directions of leaves. Segmentation was used to determine the tangential direction (TD) to retrieve the skeletons, then used relative moment and pixel directions to compute the leaf distribution.

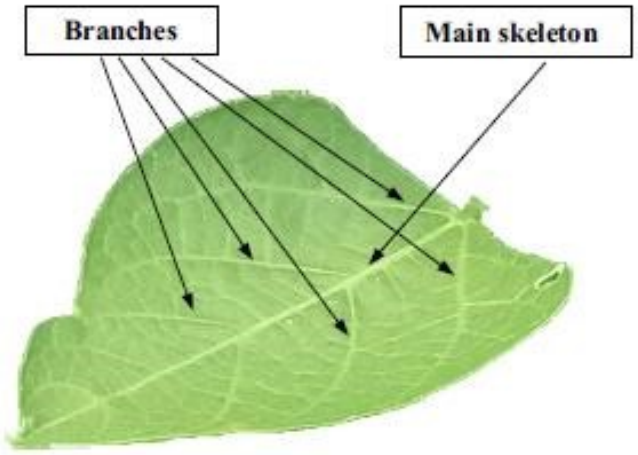


Figure 2.14. Skeleton characteristics of a leaf

The study aims to identify individual leaves automatically and to obtain leaf distribution information from images of live horticultural crops. Knowing that all types of leaves share the same skeletal characteristics, has given the study the advantage to easily identify the leaves automatically, the images can be taken from live unimpaired crops, and images can contain numerous individually identified leaves.

The study conducted 12 ordinary leaves with common features and formation belonging from 12 different agricultural plant species, and they obtained their 12 horticultural crop images from 3 different images capturing hardware: one is the wireless multimedia sensor network that comprises an interconnected camera, transmitter, and wireless network device, the other is a smartphone, and the last one is a digital camera with SD card. The data will be sent to their image center, where the skeleton features of the leaves will be extracted and identify the leaves. The purpose of the three methods is to test which device produces a clear and assures a clear shot for the leaf identification.

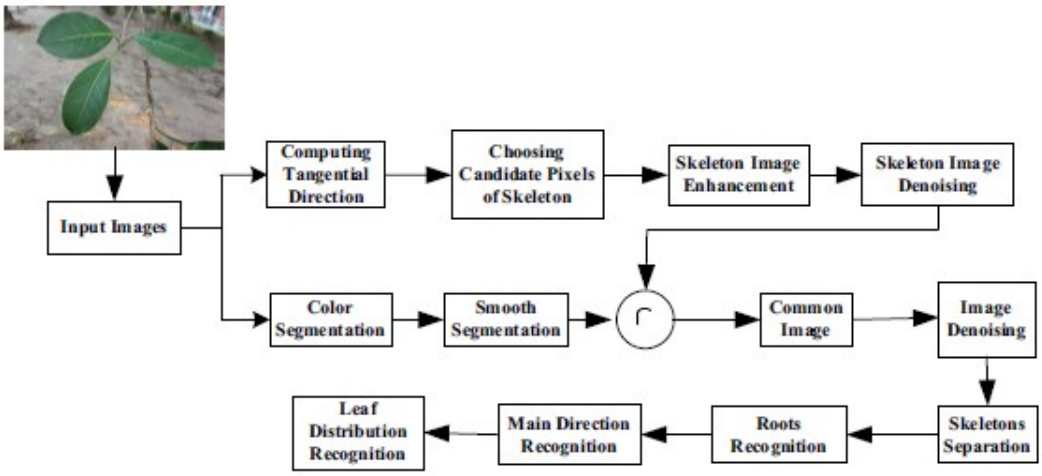


Figure 2.15. Leaf identification system

The system identification process comprises several steps, but the major and essential steps to consider are computing the tangential direction, image filtering and segmentation, and locating the root and direction to determine the distribution.

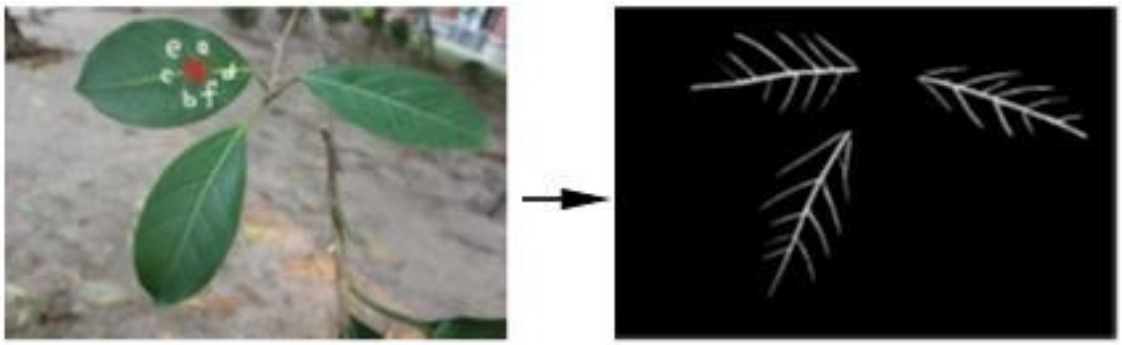


Figure 2.16. Sample candidate with tangential direction into a skeletal image

To identify the leaf skeleton, the leaf image must be illuminated by increasing the light intensity and choosing the candidate pixels within the leaf area, which is the branch since the leaf skeleton has a high luminance character. Image enhancement is then used to improve the appearance of finer details with low luminance. Afterward, unnecessary pixels scattered within the enhanced image are discarded and filtered to reduce noisy pixels. The extracting the skeletal image, the image is even more filtered through segmentation by focusing on the region of interest, which is the leaf area. As a result, the

filtered and segmented skeleton image is shown in the figure above compared to its original image.



Figure 2.17. Skeletal, root, and tangential direction recognition from an original image

Determining the skeleton does not stop the process there, the final approach would be determining the root and direction to determine the source of the distribution of leaves. As a result, the recognition of the skeletal, root, and direction are achieved and most of all, the system provided the solution to the numerous overlapping leaves, diseased or compromised leaves, obstructions, complicated backgrounds, and feature resemblance.

The LDA-based Face Recognition at a Distance using Multiple Distance Image

According to (Pan, 2013) the development of an intelligent human or facial recognition system that can handle and judge the actions of people proven to be applicable for a surveillance system. This technique for image analysis, computer vision, and pattern recognition are practical in detecting criminals captured or surveilled on camera. LDA-based face recognition algorithm for the intelligent surveillance system proposed to be appropriate for human recognitions, particularly for long-distance recognition algorithms. This study conducted an experiment to train and test 5 clear captured images with varying distances of 1m to 5m respectively. Using Linear Discriminant Analysis (LDA) of Lanczos3 with bilinear interpolation for the normalization, and Euclidean Distance to measure the similarity, the result of the experiment produced improvement in its outputs from the existing facial recognition performance of 85.8% within a short distance and 44% within long distance. The

improvement was endorsed for it improved 6.1% in short distance and 31% in long-distance producing an optimum result of 91.9% and 75.0% respectively on varying distances.

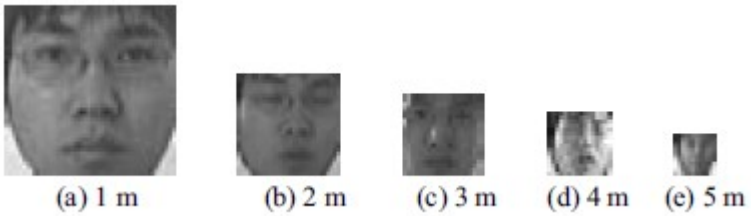


Figure 2.18. Examples of extracted face image by distance

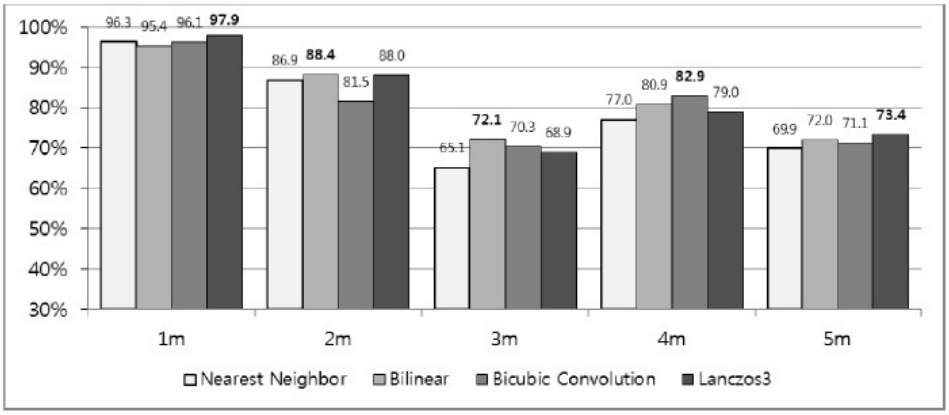


Figure 2.19. Face recognition rate according to interpolation

In the demand of the surveillance camera with special long-distance algorithm capabilities, the proposed technique using particularly Lanczos3 for short and long-distance facial recognition proved to be a game-changer. Nevertheless, the technique has the downside of requiring an extensive amount of samples of long-distance images for training to produce more accurate and consistent results, and a disadvantage of lowering recognition rate when distance further increases, which is more than 5 meters.

Research Paradigm

According to (M. Zaid, 2015) (H. Goëau, 2014), their client and most studies have stated problems involving the process of identifying trees or plants. As it has been said, it takes a long time and process of identifying tree and plant, and it is impractical for nature dwellers or the public, such a hiker, amateur stakeholders, or citizens, that has

little knowledge and identification skills and the burden of always carrying heavy guides or equipment that will help identify the species. The current process requires heavy knowledge and a field guide for detailed observation upon the feature of a tree or plant, particularly on its leaf, to decide its species and navigate its location. There are large numbers of species existing in the world and even if considering only the smaller area like the forest regions in Batangas, it would be a challenging task when it comes to manual tree inventory that requires absolute training in tree identification.

According to (Withanage, 2015) Leaves provide sufficient features to distinguish them among other species. but the identification of trees or plants using leaf images is a classic problem in digital image processing, specifically to produce accurate results. According to (L. Zhang P. W., 2016) there are very complex scenarios that make leaf identification challenging when being manipulated using image processing techniques, such as messy distributions, fuzzy edges, overlapping, and weeds. Simply, the problem is obstruction, genetic variance, and feature sharing, but they solved the problem using leaf skeleton features and estimated skeletal direction. Regardless of the negativity of the problem cases, the solution is to filter, enhance, and convert the acquired RGB or original image of the leaf into binary form for the machine algorithm to better understand its common and unique features from the leaves. Based on (Withanage, 2015) also suggested using Scale-Invariant Feature Transform (SIFT) features combined with Bag of Words (BoW) model and Support Vector Machine (SVM) classifier. This method produced a promising accurate result, tested on Android devices and server-based applications, computed from an extensive amount of image samples processed on its descriptor algorithms and feature vector classifier.

Research Hypothesis

Applying image processing on a mobile application, the process of tree or plant species identification would be automated since the server-based application provides the basic information of its accurate knowledge of the identity and with the help of the mobiles' GPS to record its real-time geographic location and distribution. Image retrieval

technologies, implemented in the applications, consider a promising direction in reducing the taxonomic gap. In addition, the application would play a major role to individuals that lack the knowledge of trees as well as experts, specifically, foresters, farmers, teachers, researchers, and botanists.

Chapter 3 METHODOLOGY

The Methodology that will be used in the study is the PPDC model. This model is composed of four phases that are closely related: Prepare, Plan, Design, Check.

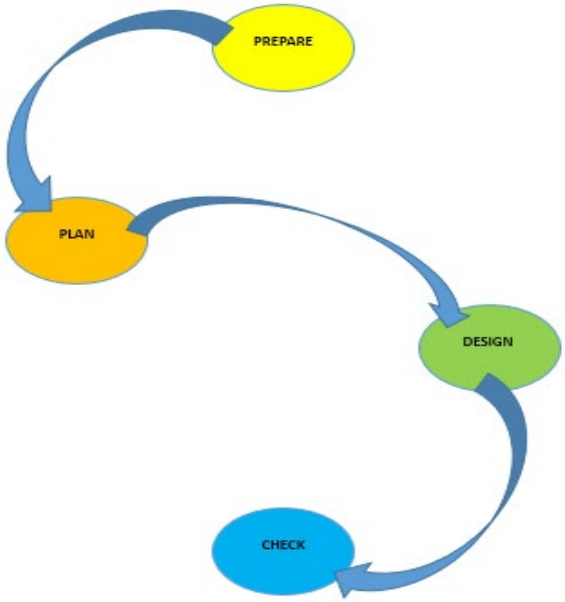


Figure 3 PPDC Model

Figure 3 shows the method that will help the researcher to gather all the needed requirements before the objectives of the study by following the different stages of the PPDC model.

Prepare Phase

Generally, in the Prepare phase, a system will develop its requirements, technology vision, and technology strategy. These must be established before going forward to ensure that the PICTRETION is as close to the objectives and requirements as possible.

3.1 Functional Analysis

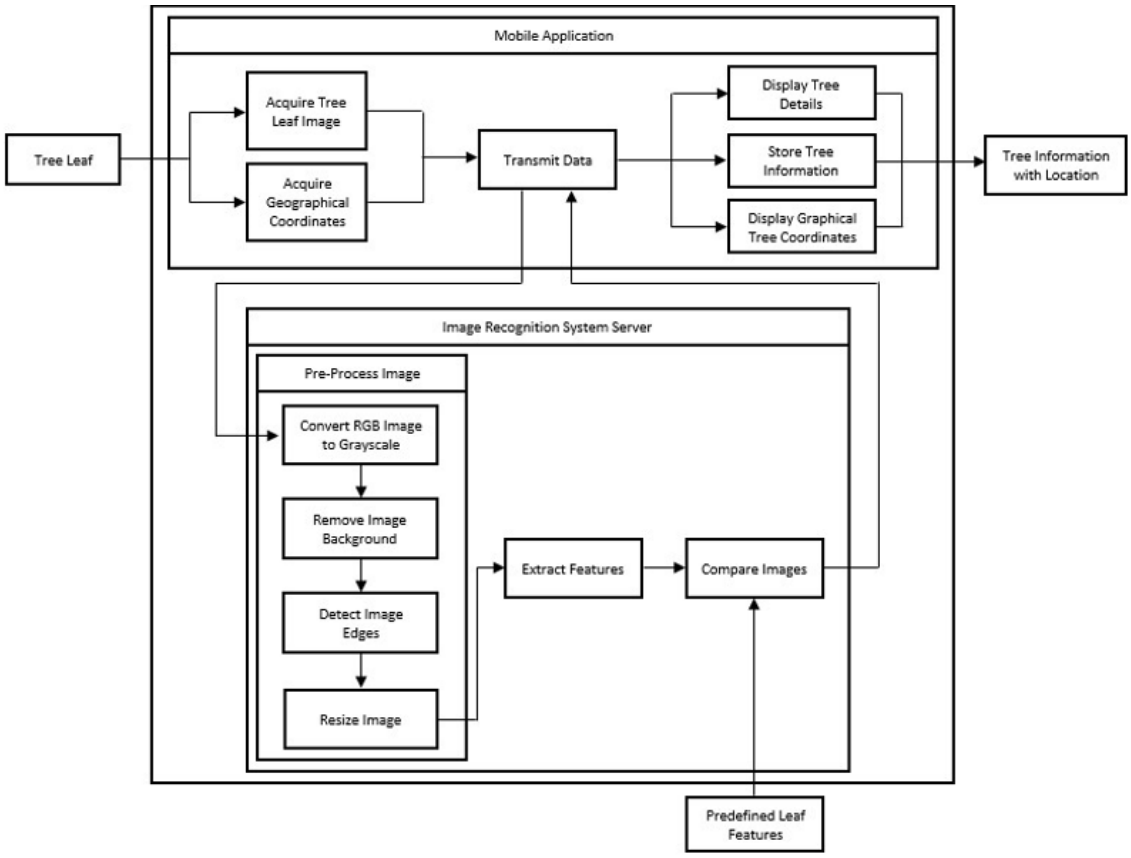
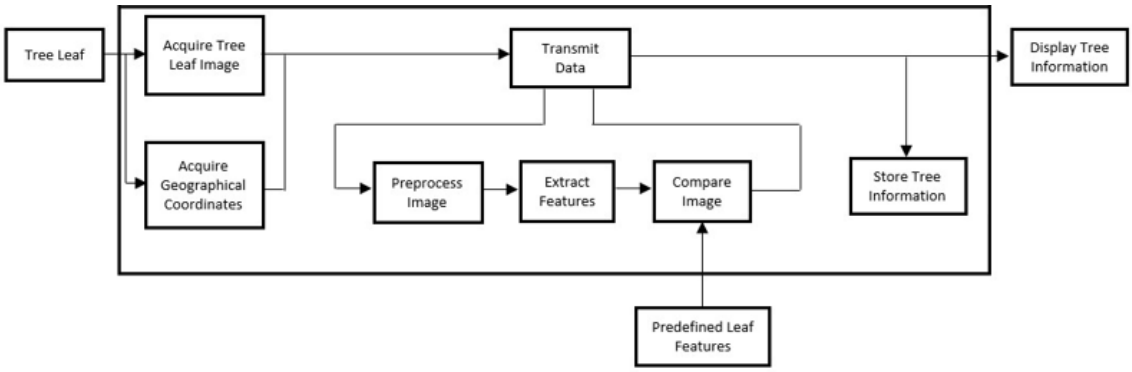


Figure 3.1. Functions and analysis

Figure 3.1 shows the different principal functions that the PICTRETION system is expected to perform. The system will start by acquiring the tree data using the android application installed in the Android device. Afterward, it would upload the data to the server where it would be pre-processed, and the features of the acquired image would be extracted. By determining the classification of tree species, the image features extracted will be compared with the pre-defined features datasets from the database to classify the

image feature. The results will be stored into the database of classified images and fed into the host device or other devices displaying the specific information of the tree and exact geographic location of the tree species visually presented on the integrated map



interface.

Figure 3.2. Glass box

Figure 3.2 illustrates the flow of the functions that the Pic-A-Tree system will perform. The diagram depicts the overall functionality of the design system wherein it shows the input, system process, predefined dataset, and output from left to right. The system represents the different primary and sub-functions of the prototype highlighting the essential roles of each function in image processing to be performed and incorporated into a single tree identifying network system application.

3.1.1 Principal Functions

- Acquire Tree Leaf Image
- Acquire Geographical Coordinates
- Preprocess Image
- Extract Image Features
- Compare Image
- Store Tree Information
- Display Tree Details

- Display Graphical Tree Coordinates

3.1.2 Secondary Functions

- Transmit Data
- Convert RGB image to Grayscale
- Remove Image Background
- Detect Image Edges
- Resize Image
-

Table 3.1. Functions and Specifications

Functions	Types	Specifications	Classifications
Acquire Tree Leaf Image	Principal	The leaf image will be acquired using the built-in camera in the selected phone at above 16mp	Performance
Acquire Geographical Coordinates	Principal	The geographic coordinates that include both longitude and latitude will be acquired using the built-in G.P.S technology feature of smartphones.	Performance
Transmit Data	Secondary	The data that was transmitted through the internet will send and receive data to and from the dedicated server	Performance
Preprocess Image	Principal	The raw image will undergo the pre-processing technique of image processing utilizing grayscaling, segmenting, extracting edges, and resizing an image	Procedural
Extract Features	Principal	The system will extract the different object-based unvarying keypoints and descriptors set as features deconstructed from the pre-processed image	Procedural
Compare Image Testing	Principal	The system will compare the processed image to different pre-defined images that will search for matching features of species identities.	Procedural
Store Tree Information	Principal	The information acquired from the transmitted data would be arranged and stored in the Pic-ATree database provided by the prescribed client-server.	Procedural
Convert RGB Image to Grayscale	Secondary	The acquired leaf image will be converted from a colored image to a grayscale image by transforming colored pixels within an RGB image to black and white pixels	Procedural
Remove Image Background	Secondary	The image converted into grayscale are now apt for image background removal that segments or masks the image separating the ROI from the unwanted regions or modifying the background and foreground as binary representing black and white	Procedural

Detect Image Edge	Secondary	After segmentation, the function highlights and contours the potential features from the segmented image	Procedural

Table 3.1. Functions and Specifications (continuation)

Function	Type	Specification	Classification
Resize Image	Secondary	The modified detected image will use the interpolation method to enlarge or increase the resolution of the modified image maintaining quality as of the original image	Procedural
Display Tree Details	Principal	The classified image taken after the image comparison function will display the tree information to devices transmitted from the network server	Performance
Display Graphical Tree Coordinates	Principal	With the tree information, the tree geographic location or coordinates will be visual by presented on the integrated mapping application programming interface (API) of the application	Performance

Table 3.1 shows the different functions and specifications of the proposed project. The table above is already self-explanatory indicating the functions, specification type, function description or specification, and classification. In terms of specification type, each function is classified whether principal or secondary. Principal pertains to the parent or main function that will perform within the design function, while secondary are sub-functions that are part of the main function or related to the parent function that can consist of two or more functions. The function of transmission of data specifies that it is a secondary function, but not a sub-function, describing that it is not part of the procedural function of the system, somehow system performance that normally is executable during connectivity. While the considered sub-functions, those sub-functions are grayscale conversion, image background removal or segmentation, edge detection, and resizing of the image, which are coherent sub-functions of the pre-processing image main function. Also, the different functions are categorized as either procedural or performance.

Procedural specifications explaining data processing are dependent on other functions, rather than containing requisites that require different sub-functions to be successful. As for performance specification, it performs independently with simple and particular operations defined as performance level functions that must be demonstrated for it to be successful.

3.2 Design Alternatives

Table 3.2. Design Space

Functions	Engineering Standard	Standard Description	Means		
Acquire Tree Leaf Image	ISO / IEC TS 20071	Guides various aspects of the user interface of applications that scan visual information that is used directly by humans.	Smartphone 1	Smartphone 2	Smartphone 3
Acquire Geographical Coordinates	ISO 6709	Applicable to the interchange of coordinates describing geographic point location. It specifies the representation of coordinates, including latitude and longitude.			
Transmit Data	ISO/IEC JTC 1/SC 6	Standardization in the field of telecommunications dealing with the exchange of information between systems.	WIFI		
Preprocess Image	ISO/IEC 15948	Specifies a data stream and an associated file format compressed individual computer graphics image transmitted across the Internet.	OpenCV	MathLab	Python
Compare Image Testing	ISO/IEC 29170	Evaluation and grading of a light coding system used for display systems but is independent of the technology. This procedure measures whether an observer can distinguish between an uncompressed reference and the reconstructed image to a pre-determined, statistically meaningful level.	k-Nearest Neighbor (kNN)	Support Vector Machine (SVM)	Naive Bayes/Bayesian
Store Tree Information	ISO/IEC 13249	Defines several packages of generic data types common to various kinds of data used in application areas.	SQLite	MySQL	Microsoft SQL Server
Extract Features	ISO/IEC 15948	Specifies a DataStream and an associated file format, compressed individual computer graphics image transmitted across the Internet. Indexed-color, greyscale, and TrueColor images are supported, with optional transparency. Sample depths range from 1 to 16 bits. Providing both full file integrity checking and simple detection of common transmission errors	Scale Invariant Feature Transform (SIFT) Descriptor	Histogram of Oriented Gradients (HOG) Descriptor	Speeded-up Robust Features (SURF) Descriptor
Convert RGB image to Grayscale			Luminance	Intensity	Luster
Remove Image Background			Thresholding	Edge-Based Masking	Region Growing Method
Detect Image Edges			Sobel Operator	Canny Edge Detector	Prewitt Edge Detector
Resize Image			Edge Classification	Bilinear	Bicubic

Table 3.2 shows the engineering standards, descriptions, and means of different functions of the research. In Acquire tree leaf image, the User interface component accessibility (ISO / IEC TS 20071) where it guides various aspects of the user interface of applications that scan visual information. In acquiring geographical coordinates, the necessary standard for the representation of geographic point location through coordinates (ISO 6709) applies to the interchange of coordinates describing geographic point location. It includes latitude and longitude wherein among 3 dedicated Androids smartphones integrated with GPS technology will be the candidates for mean selection. The highest quality camera resolution of the smartphone will be the primary focus upon recognizing images at a limited scale. The following constraint occurs on the price if it reaches near or beyond the allowance allotted for the smartphone device or the entire project design, which will be least considered. In Transmit Data to Client Server, Telecommunications and information exchange between systems (ISO/IEC JTC 1/SC 6) is telecommunication dealing with the exchange of information between systems. Since 4G, 3G, and other network technologies are incomparable and insignificant compare with reliable Wi-Fi technology, it is safe to presume that Wi-Fi can perform transmission of data even with obstructive conditions or unexpected events that may result from catastrophic transmission failure, hence, overpowering other network technology. In comparison, image testing, advanced image coding, and evaluation (ISO/IEC 29170) this procedure measure whether an observer can distinguish between an uncompressed reference and the reconstructed image to a pre-determined, statistically meaningful level. In-store tree information, database languages (ISO/IEC 13249) define several packages of generic data types common to various kinds of data used in application areas. In extract features, convert RGB image to grayscale, remove image background, detect image edges, and resize image under pre-processing, computer graphics and image processing (ISO/IEC 15948) specifies the data stream and an associated file format, compressed individual computer graphics image transmitted across the Internet. The means selection of the preprocessing function will be based on the platform that can accommodate all procedural sub-function of the pre-processing stage.

Plan Phase

During the Plan Phase, the researcher performs a plan on how the system will work based on the objectives, include here the flowchart of the system, how the selection process for leaf image acquisition**,** geographical location acquisition, and how the image will process using the system.

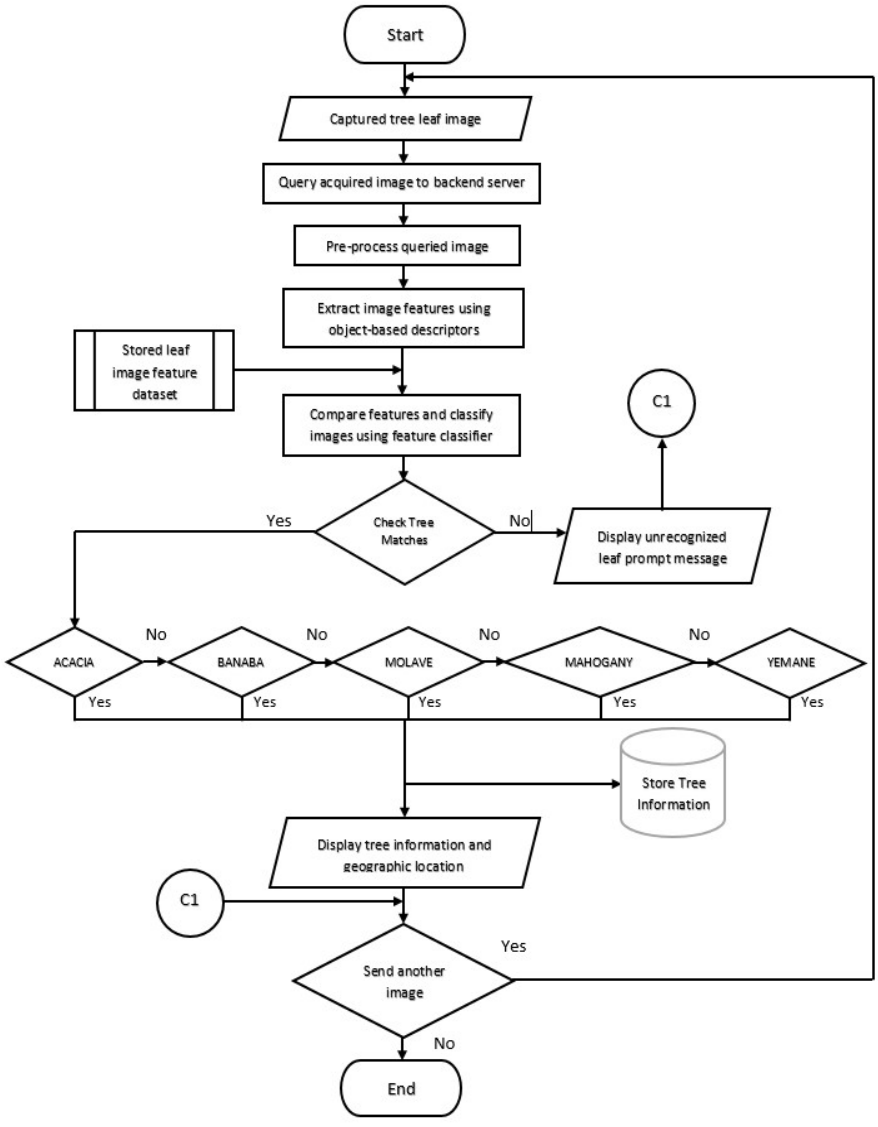


Figure 3.4. System flowchart

Figure 3.4 In the flowchart above, the proponents used various symbols which represent the type of function/process our system utilizes. From the start, the user will input the image of the tree leaf. After which, it will be sent to the webserver for the backend server to receive and where it will be processed using Python image processing. The first step in image processing is through preprocessing where the image will be simplified by gray image conversion, image segmentation, edge detection, and image resize. The simplified image will be processed in the feature descriptor algorithm extracting its potential feature key points as data. The features will be the essential data for the machine learning stage used to compare the trained data with the raw data. With this image recognition system process, the system will able to classify the species of the leaf and will save the information back to the webserver consisting of the tree species with its geographic location. The results will be observed externally on the Android application and API map interface integrated into the system comprising the identified tree species.

3.3 Selection Process

The proponents have chosen the best-of-class method in the selection of means that will best fit the respective functions. For the selection process, it is important to consider the design objectives and the ranking of each means will be based on the priority weight utilizing scoring each design objective. To measure the proposed design credibility, the proponents based the evaluation of each means according to the priority level with the respective scores, i.e. 1 as the first option attains 100% of the score, 2 as the second option attains 50% of the score, and 3 as the third option can attain 25% of the score. However, if a certain case involves both means produced similar scores for the criteria, the proponents will decide, which means to select based on the significant weight of the design objectives to be performed by the system. In this case, the accuracy and reliability are considered first and second critical respectively, followed by the response time and/or consistency considered next critical if the first 2 foremost objectives were not prescribed. The selected mean with the highest score will serve as first option for the function. The selection for the priority of each means is based on testing, verification, facts and specification of the means that define each design objective, wherein the objectives that

were considered are accuracy, reliability, responsiveness, consistency, and user-friendliness.

Table 3.3. Selection Process for Leaf Image Acquisition

Acquire Tree Leaf Image							
Priority Score Attainment:		**1st = 100%**		**2nd = 50%**		**3rd = 25%**	
Design Constraints (C) and Objectives (O)	**Priority Weight**	**Smartphone 1**		**Smartphone 2**		**Smartphone 3**	
		Priority	**Score**	**Priority**	**Score**	**Priority**	**Score**
O: Accurate							
O: Reliability							
O: Responsiveness							
O: Consistency							
O: User-Friendly							
Total Score							

Table 3.3 shows the selection process function of acquiring a tree leaf image. The possible means for this function are Smartphone 1, Smartphone 2, and Smartphone 3. Based on the required specification for selecting an adequate means for this function, considering the pixel resolution that each means produces.

Table 3.4. Selection Process for Geographic Location Acquisition

Acquire Geographic Coordinates							
Priority Score Attainment:		**1st = 100%**		**2nd = 50%**		**3rd = 25%**	
Design Constraints (C) and Objectives (O)	**Priority Weight**	**Smartphone 1**		**Smartphone 2**		**Smartphone 3**	
		Priority	**Score**	**Priority**	**Score**	**Priority**	**Score**
C: Geographical Coordinates							
O: Responsiveness							
O: User-Friendly							
Total Score							

Table 3.4 shows the selection process function of acquiring the geographic location of the tree leaf. The possible means for this function are Sony Xperia XA Ultra, Xiaomi Redmi 5A and Cherry Mobile Flare S6 with provided GPS technology in the prescribed devices included namely with A-GPS featuring GLONASS, A-GPS exclusively, and A-GPS featuring GLONASS and BDS respectively. Based on the required specification for selecting an adequate means for this function, the G.P.S capability of Xiaomi Redmi 5A can be promising considering its GPS feature uniquely supports BDS, using wide area geo-satellites mainly covering Asia, and GLONASS, covering regions at high latitude (north and south continents). It is above the expected values that were presented in the specification thus making Xiaomi Redmi 5A GPS technology the option suitable for the function.

Table 3.5. Selection Process for Image Processing Platforms Capable of Preprocessing

Pre-process Image							
Priority Score Attainment:		**1st = 100%**		**2nd = 50%**		**3rd = 25%**	
Design Constraints (C) and Objectives (O)	**Priority Weight**	**OpenCV-Python**		**MathLab**		**Python**	
		Priority	**Score**	**Priority**	**Score**	**Priority**	**Score**
O: Responsiveness							
O: User-Friendly							
Total Score							

Table 3.5 shows the selection process for pre-processing using renowned computer vision or image recognition platforms. The possible means for the function are OpenCV-Python, MathLab, and Python. The following platforms are popular for commercial and education since all platforms provide all the necessary tools, libraries, functions, and other resources to perform procedural algorithms needed and apt for image processing. The main difference is that MathLab consumes much RAM and has many powerful matrix libraries due to its abundant background supportive features during execution. In this case, MathLab computational complex and robust than OpenCV with Python combined. However, the downside of MathLab is that it extremely costs greatly more or less $2000 expensive for commercial use and 3x slower runtime than OpenCV-Python, next with

Python exclusively, is freeware, open-source cross-platform. Despite the bias, MathLab, followed by Python, is more practical and user-friendly for beginners and intermediate developers everywhere than OpenCV since it provides easier coding of C/C++ for debugging, visuals, and supportive documentation to educate or help developers learn by themselves the unique functions of codes and features, making Mathlab an educative platform particularly for image processing. Therefore, the specification presented makes OpenCV-Python platforms the option suitable for the function.

Table 3.6. Selection Process for Image Grayscale Conversion

Convert RGB Image to Grayscale							
Priority Score Attainment:		**1st = 100%**		**2nd = 50%**		**3rd = 25%**	
Design Constraints (C) and Objectives (O)	**Priority Weight**	**Luminance**		**Intensity**		**Luster**	
		Priority	**Score**	**Priority**	**Score**	**Priority**	**Score**
O: Accuracy							
O: Reliability							
O: Responsiveness							
O: Consistency							
Total Score							

Table 3.6 shows the selection process function of converting the RGB image into a grayscale format. The viable algorithms best to transform a colored image into a grayscale format are luminance, intensity (average), and luster. According to expert developers, the luminance algorithm is more accurate for image processing since it uses the weighting method of RGB that is perceivable by the common eyes, compared to the intensity that uses the mean method, which cannot. While for luster, it focuses on HLS format (Hue, Lightness, Saturation) that reduces image contrast more appropriate for computer graphic designing. In this case, it makes the first option suitably reliable for computer vision. Since all algorithms run using OpenCV-Python or Mathlab preset functions, all algorithms are considered responsive, but the overall intensity method has a more responsive execution due to its simple computational structure, but likely inconsistent in producing optimum results due to conventional, unelaborated method compared to the

other methods. It is above the expected values that were presented in the specification thus making the luminance method the option suitable for the function.

Table 3.7. Selection Process for Image Background Removal

Remove Image Background							
Priority Score Attainment:		**1st = 100%**		**2nd = 50%**		**3rd = 25%**	
Design Constraints (C) and Objectives (O)	**Priority Weight**	**Thresholding**		**Edge-Based Masking**		**Region Growing Method**	
		Priority	**Score**	**Priority**	**Score**	**Priority**	**Score**
O: Accuracy							
O: Reliability							
O: Responsiveness							
O: Consistency							
Total Score							

Table 3.7 shows the selection process function of removing the image background. The viable algorithms for segmenting images or obtaining the Region of Interest (ROI) are the thresholding, masking, and region-growing methods. In a long line of image segmentation, the thresholding method is the simplest form of a method of recognizing and segments the ROI with the background and unnecessary pixels. Thresholding provides speed from its adequate computational process, and the next best thing is edge-based masking but increases in computational complexity as the operators' job or size of the image increases.

Regardless of its speed due to tedious massive computation of image rendering, Region Growing Method (RGM) provides accurate results and reliability from the Seeded region classification providing robust and rapidness of generating fitting parameters suitable for large scale images. Furthermore, RGM and edge-based masking execute their respective process with improved noise immunity, compared to thresholding, which is prone to noise. It is above the expected values that were presented in the specification thus making the region growing method the option suitable for the function.

Selection Process for Image Edge Detection

Detect Image Edges							
Priority Score Attainment:		**1st = 100%**		**2nd = 50%**		**3rd = 25%**	
Design Constraints (C) and Objectives (O)	**Priority Weight**	**Sobel Operator**		**Canny Edge Detector**		**Prewitt Edge Detector**	
		Priority	**Score**	**Priority**	**Score**	**Priority**	**Score**
O: Accuracy							
O: Reliability							
O: Responsiveness							
O: Consistency							
Total Score							

Table 3.8 shows the selection process function of detecting the image edges. The viable detector algorithms best used for image edge detection are Sobel, Canny, and Prewitt. Sobel is the simplest of them all, while Prewitt, being the second-best in process time and workspace, and finally the most complex of all, the canny consumes more process time. Canny and Sobel are considered accurate due to their adjacent level of ideal thresholding for detecting image edges, but canny is more robust for detecting false images. For reliability, the noise of each detector was tested and proven that canny and Prewitt are more stable with noise sensitivity. Canny, being complex, has a downside of having less process time compared with the two, but in terms of consistency, canny is off the charts since it can perform edge detection effectively to any kind of captured image from satellite imagery to short distance imagery due to its reduced chances of false positives. It is above the expected values that were presented in the specification thus making the canny edge detector algorithm the option suitable for the function.

Selection Process for Image Size Modification

Resize Image							
Priority Score Attainment:		**1st = 100%**		**2nd = 50%**		**3rd = 25%**	
Design Constraints (C) and Objectives (O)	**Priority Weight**	**Edge Classification**		**Bilinear**		**Bicubic**	
		Priority	**Score**	**Priority**	**Score**	**Priority**	**Score**
O: Accuracy							
O: Reliability							
O: Responsiveness							
O: Consistency							
Total Score							

Table 3.9 shows the selection process function of resizing the modified image. The viable detector algorithms best used for image resizing are edge classification method, bilinear and bicubic interpolation. Bilinear and bicubic are the traditional image resizing algorithms for image preprocessing. Practically speaking, bilinear is more applicable for downsizing images, and bicubic is for upsizing, but both methods increase the pixel amount of the original for interpolation from extensive and complex computation and counterparts each other even if their results are slightly equivalent. However, the edge classification method competes with the traditional methods. This method focuses on edge-based details for interpolation as it accurately and consistently measures the texture, pixel size, and edges of the original image. Compared to the two methods, the edge classification method is more reliable in upscaling by maintaining its edge details than bilinear and bicubic producing blurry and less effective edge details. Despite the advantage, edge classification is computationally complex than the other two consuming more process time, but more consistent in image resizing due to its robust edge detecting algorithm. It is above the expected values that were presented in the specification thus making the edge classification the option suitable for the function.

Selection Process for Image Feature Extraction

Extract Image Features							
Priority Score Attainment:		**1st = 100%**		**2nd = 50%**		**3rd = 25%**	
Design Constraints (C) and Objectives (O)	**Priority Weight**	**Scale-Invariant Feature Transform (SIFT)**		**Histogram of Oriented Gradients (HOG)**		**Speeded-up Robust Feature (SURF)**	
		Priority	**Score**	**Priority**	**Score**	**Priority**	**Score**
O: Accuracy							
O: Reliability							
O: Responsiveness							
O: Consistency							
Total Score							

Table 3.10 shows the selection process function of extracting features. The following object-based features descriptors used are SIFT, HOG, and SURF. As a default, the HOG is discarded since it falls behind 50% of the accuracy and precision (reliability) level compared with the other two. Technically, the SIFT descriptor is more robust compared to SURF due to its unique features invariant to illumination, perspective, and scaling, producing high accuracy and identity precision. Unlike SIFT and HOG, SURF is much efficient in processing feature data due to its simple computation and algorithm making it uniquely responsive to any descriptors. In addition, SIFT has reduced the level of noise or nearly noise immune proving it to be consistent in performance. Practically, SIFT is the best choice for machine learning applications due to its flexibility in obtaining object features for image processing. It is above the expected values that were presented in the specification thus making the SIFT descriptor algorithm the option suitable for the function.

Table 3.11. Selection Process for Image Feature Comparison and Classification

Compare Image Features							
Priority Score Attainment:		**1st = 100%**		**2nd = 50%**		**3rd = 25%**	
Design Constraints (C) and Objectives (O)	**Priority Weight**	**k-Nearest Neighbor (kNN)**		**Support Vector Machine (SVM)**		**Naïve Bayes/Bayesian Nearest Neighbor**	
		Priority	**Score**	**Priority**	**Score**	**Priority**	**Score**
O: Accuracy							
O: Reliability							
O: Responsiveness							
O: Consistency							
Total Score							

Table 3.11 shows the selection process function of comparing image features. The following means are common classifiers namely stated as k-NN, SVM, and Naïve Bayes technique. The k-NN is the most common and simple supervised learning algorithm whereby classifying the locating nearest trained datasets with sample data, while Naïve Bayes is a sophisticated learning method where classification depends on its developed theorem. Compared with k-NN, SVM has a slightly higher accuracy rating making it a good alternative competition for classifiers, but still likely to have the same result. Based on its capabilities, SVM has a lot to offer or more reliable since it can be useful for classifying massive datasets, unlike k-NN or Naïve Bayes which have limitations in classification. For response time, SVM is slower than the two due to its robust classification method but is more consistent in approaching accurate results based on the generated sample hyperplanes for classification. It is above the expected values that were presented in the specification thus making the SVM classifier algorithm the option suitable for the function.

Table 3.12. Selection Process for Management of Database in Server

Store Tree Information							
Priority Score Attainment:		1st = 100%		2nd = 50%		3rd = 25%	
Design Constraints (C) and Objectives (O)	Priority Weight	SQLite		MySQL		MS SQL Server	
		Priority	Score	Priority	Score	Priority	Score
O: Responsiveness							
O: User-Friendly							
Total Score							

Table 3.12 shows the selection process for the functional management of the database in the server. The possible means for this function are SQLite, MySQL, and Microsoft SQL Server. It is based on the required specification for selecting an adequate means for this function, considering the characteristics on the table above and the ability of each software to manage data of an Android application within a relational database. SQLite is a choice considering its ability to connect towards a specific line of data and the wide of versions it covers on android since it supports solely on C language with similar query statements of MySQL. SQLite is also considered lightweight, easy to use, and readily and natively available within the android application. However, other relational databases, such as MySQL, cater to more features that as data security, backup and recovery, data migration and replication, and much more. It remains a renowned and growing database management tool for huge companies and enterprises based on its performance and specification, but the downside is due to regular membership to maintain system operation for remote connectivity, whereas MS SQL is a freeware database management system that provides similar specs and features with MySQL except has reduced simplicity particularly for small-scale application since by far the UI setup of Microsoft applications are developed for business class system software. It is above the expected values that were presented in the specification thus making the SQLite database provider the option suitable for the function.

Table 3.14. Selection Process for Display of Geographical Tree Coordinates

Display Geographic Information							
Priority Score Attainment:		**1st = 100%**		**2nd = 50%**		**3rd = 25%**	
Design Constraints (C) and Objectives (O)	**Priority Weight**	**Google Maps**		**Mapfit**		**TomTom**	
		Priority	**Score**	**Priority**	**Score**	**Priority**	**Score**
O: Accuracy							
C: Identified Trees							
O: Responsiveness							
O: User-Friendly							
Total Score							

Table 3.14 shows the selection process for the function of displaying geographical tree coordinates. The possible means for this function are Google Maps, Mapfit, and TomTom. Based on the required specification for selecting an adequate means for this function, considering the characteristics on the table above. All three maps have their measure of accuracy, which they claim on their website to be a 99% coverage of the world. Hence, they are all equal inaccuracy, but Google Maps has proven to be accurate due to the number supportive of GPS modules next to TomTom. Google maps support the multiple numbers of G.P.S modules that are integrated with different phones which include but are not limited to A-GPS, GLONASS, and BDS. Google maps are more accurate and reliable in terms that it has a bigger server than the other two services. Mapfit provided a more user-friendly experience in an android setting rather than Google, as it has features that allow the user to remove clutter from the screen and mark their entrances and exits on an establishment. It is also based on location access of Big Data making real-time process consistently effective for updates per minute. It is above the expected values that were presented in the specification thus making the Google maps the option suitable for the function.

3.4 Function Means Matrix

Table 3.15. Function Means Matrix

Functions	**Means**		
Acquire Leaf Image			
Acquire Geographic Location			
Transmit Data			
Pre-Process Image			
Convert RGB Image to Grayscale			
Remove Image Background			
Detect Image Edges			
Resize Image			
Extract Features			
Compare Image Features			
Store Tree Information			
Display Tree Details			
Display Geographical Tree Coordinates			

Table 3.15 shows the summary of the design space for the selected means for each respective function. For the acquiring input data. The OpenCV-Python used has been utilized as the platform/s for the implementation of image recognition procedural tasks. As for the image recognition procedures, the pre-processing stage will undergo the luminance method for grayscale conversion, region growing method for segmentation, canny edge detector for edge detection, and bicubic interpolation for resizing the image. Combining these sub-functions will create a stable, accurate, and robust pre-processing in preparation for extract features using SIFT feature descriptors and SVM classifier to

compare the image with predefined feature dataset stored in the feature database. For the function of storing data within the database server, it proved that SQLite allows the user to access additional functions within the prescribed application with higher storage capacity compared to other counterparts. Lastly, the output will be the results of the identified tree and detailed information of its background where results will be posted in the application on Android devices and specific location on the dedicated API map module not likely to be Google maps.

3.5 Mathematical Models

3.5.1 Scale Invariant Feature Transform (SIFT) Keypoints

According to the 2004 paper of David Lowe, *Distinctive Image Features from ScaleInvariant Keypoints*, the approach for obtaining SIFT feature key points requires 4 major stages of computation to generate scale-invariant features on multiple possible scales and angles wherein the 4 major stages are: **scale-space extrema detection**, **keypoint localization**, **orientation assignment**, and **keypoint descriptor**.

Scale-Space Extrema Detection

$$\begin{aligned} D(x, y, \sigma) &= (G(x, y, k\sigma) - G(x, y, \sigma)) * I(x, y) \\ &= L(x, y, k\sigma) - L(x, y, \sigma). \end{aligned}$$

where $\sigma(k\sigma) -$ *scaling parameter of the Laplacian of Gaussian*$(original\ scale)$,
$L(x, y, k\sigma) = G(x, y, k\sigma) * I(x, y)$,
and, $L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$.

Keypoint Localization

$$D(\mathbf{x}) = D + \frac{\partial D}{\partial \mathbf{x}}^T \mathbf{x} + \frac{1}{2}\mathbf{x}^\mathbf{T}\frac{\partial^2 D}{\partial \mathbf{x}^2}\mathbf{x}$$

where D *and derivatives are evaluated through the sample point,* $x = (x, y, \sigma)^T$ *as the offset point.*
And

$$\hat{\mathbf{x}} = -\frac{\partial^2 D}{\partial \mathbf{x}^2}^{-1} \frac{\partial D}{\partial \mathbf{x}}.$$

As the final function value for the extrema to reduce noises and reject unstable extremes, substituting equation 3.2 and 3.3 will obtain the formula given,

$$D(\hat{\mathbf{x}}) = D + \frac{1}{2} \frac{\partial D}{\partial \mathbf{x}}^T \hat{\mathbf{x}}.$$

Orientation Assignment

The scales of sample key points are computed in a scale-invariant manner with an applied selection of Gaussian smoothed image L(x,y), with the nearest scale, but precomputed with pixel differences.

$$m(x,y) = \sqrt{(L(x+1,y) - L(x-1,y))^2 + (L(x,y+1) - L(x,y-1))^2}$$

$$\theta(x,y) = \tan^{-1}((L(x,y+1) - L(x,y-1))/(L(x+1,y) - L(x-1,y)))$$

where $L(x, y)$ *is the scale,*
$m(x, y)$ *the gradient magnitude,*
and $\theta(x, y)$ *the orientation*

Keypoint Descriptor

The key point descriptor allows a significant shift in gradient positions by creating orientation of histograms over 4x4 sample regions computed from a 16x16 sample array. To avoid boundary affecting the descriptor abruptly changing from one histogram to another, it uses trilinear interpolation to distribute the value of each gradient sample into histogram bins. Therefore, the formula to compute for adjacent histogram bins is,

$$Trilinear\ Interpolation = Sample\ Entry\ bin * (1 - d)$$

where d *the distance of the sample from central value of the bin as measured in units of the histogram bin spacing*

Equation 3.1 briefly explains the first stage of computation wherein it searches all possible scales and image locations, implemented efficiently by the difference-of-gaussian (DoG) function to identify scale-invariant and orientation keypoints. Equation 3.4 explains the computation to select candidates of location and scale that key points can be measured for stability. Equations 3.2 and 3.3 were substituted to form the formula.

Equations 3.5 and 3.6 generate various orientations assigned to each keypoint location based on image gradient vectors.

Equation 3.7 computes local image gradients to be measured in a particular region of a scale at a key point. The transformation allows significant levels of local shape distortion and change in illumination.

3.5.2 Support Vector Machine (SVM)

According to the paper of T. Hastie, R. Tibshirani, and J.H. Friedman, Section 4.5 (Separating Hyperplanes) from *Elements of Statistical Learning*, the basic formulation for generating hyperplanes is given by [52]:

$$f(x) = \beta_0 + \beta^T x$$

where β *as the weight vector and* β_o *as the bias*,

However, optimum hyperplanes can also be represented in an infinite number by scaling β and β_o, in a conventional matter of representing support vectors of a hyperplane, the one applied is,

$$|\beta_0 + \beta^T x| = 1$$

where x *represents the training exampes closest to the hyperplane*,

Given the distance of support vectors between a sample point of x and a hyperplane (β, β_o)

$$\text{distance}_{\text{support vectors}} = \frac{|\beta_0 + \beta^T x|}{\|\beta\|} = \frac{1}{\|\beta\|}.$$

and recalling the denotation of the margin, $M = \frac{2}{\|\beta\|}$, with the problem of maximizing the margin has a significant equivalent with the problem of minimizing the function L(β) subject in some constraint levels. Therefore, the hyperplane can classify correctly all the trained examples χ_i, by computing the canonical hyperplane formula

$$\min_{\beta,\beta_0} L(\beta) = \frac{1}{2}\|\beta\|^2 \text{ subject to } y_i(\beta^T x_i + \beta_0) \geq 1 \ \forall i.$$

where y_i represents each of the labels of the training example

With this method, the problem of the Lagrangian optimization can be solved using the Lagrange multipliers of equation 3.11 to obtain the weight vector β and the bias β_0 of the optimal hyperplane.

3.6 Design Phase

The design phase where how the graphical user interface (GUI) gives a user-friendly experience a how the system will interact with the user, The following figures below show on how the GUI looks like.

3.6.1 Software Application Design



Figure 3.5. Application main menu

Figure 3.5 shows the main menu screen of the application. The main menu of the application displays three icons which act as buttons to move to the different functions of the application. The first icon on the left shows a camera because it is the main method to classify the tree. Clicking the button would lead the user to take the picture of the tree leaf and the application would then try to classify the tree. The second icon on the top shows a map because it would lead the user towards the map section of the application which shows the location of the identified trees via a map. The third icon on the right shows trees because it would lead the user to the main page where all the listed trees are shown.

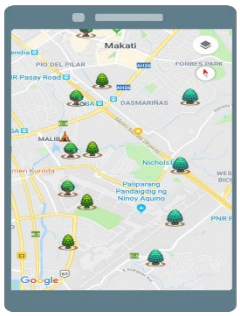


Figure 3.6. Map interface

Figure 3.6 shows the map of the application. The screenshot shows the map of the application. The tree icons symbolize the trees registered in the application.



Figure 3.7. Map interface with tree information

Figure 3.7 shows the map of the application when the icon is clicked, the screenshot shows the map of the application. The tree icons symbolize the trees registered in the application. When clicked, the application would show a quick view of the details of the tree.

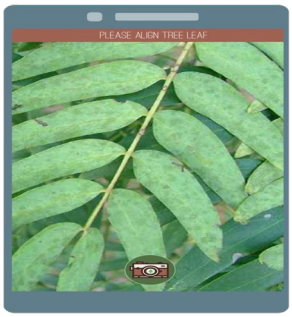


Figure 3.8. Captured tree leaf image using Android smartphone

Figure 3.8 shows the application when taking a picture of the tree leaf. The user would be reminded to align the tree leaf after which clicking the camera icon on the center would take the picture of the tree leaf and the geographic coordinates of the phone at the same time. Once the picture is uploaded, they would be in a loading screen after which, the picture is identified, they would be asked to confirm if the tree is identified correctly.

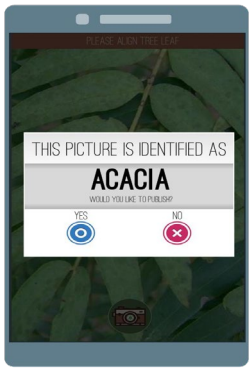


Figure 3.9. Classified leaf image notification

Figure 3.9 shows the application when confirming if the tree was identified correctly. The user would be shown the screenshot if the application would be able to successfully

find traits of a tree. The user would then be asked to confirm whether the tree is identified correctly or not. If the user confirms that the tree is correctly identified it would be placed on the tree database and the map based on the coordinates obtained through the GPS of the phone.

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Figure 3.10. Tree database menu

Figure 3.10 shows the trees that can be selected when inside the tree database. The menu is divided into 5 species which will be identified by the application. When the user selects a specific species they would be directed towards a page that lists all the registered trees for that specific species.



Figure 3.11. Tree information inside a database

Figure 3.11 shows the list of registered trees in the application The screenshot shows the registered trees including the Id number the date the tree was registered. The database also shows the coordinates of the tree and comes with a search function for the trees.



Figure 3.12. Application front page

Figure 3.12 shows the front page of the application. The front page of the application displays the main logo of the application and the name of the said application. This screen would also act as a buffer or loading screen when the application is loading data.

3.6.2 System Block Diagram

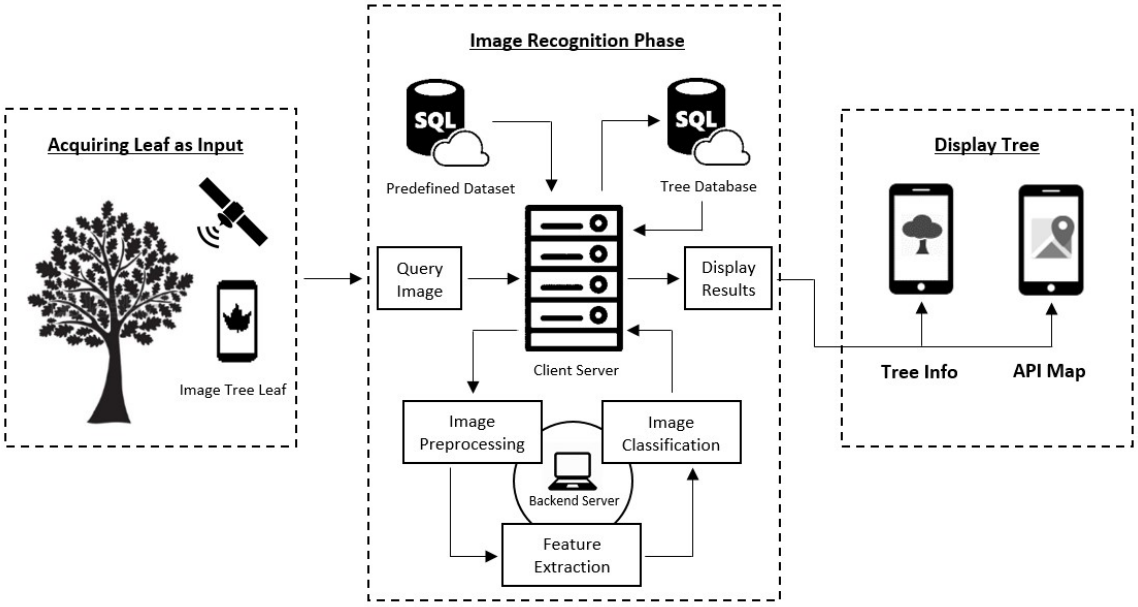


Figure 3.13. System block diagram

Figure 3.13 shows the flow of the input of the design project output. The tree leaf serves as the input parameter that will be queried to the server after being acquired using a smartphone camera and will be fed afterward the image pre-processing scheme where it will become a modified image ready for image feature extraction. The features extracted will serve as the parameters to compare it with a trained or predefined dataset and classify the given leaf features to identify the species of the tree. The output will be displayed within the application on the android devices where the tree information will be saved in a table form and tree geographic location will be plotted in the prescribed API map module.

3.6.3 Consideration Rubrics

Table 3.16. Design Consideration Table

Design Consideration	Project Application
A. Health and Safety	To recognize the importance and population of the tree.
B. Economic	Designate the proposed budget for the device.
C. Environment	Determine the number of trees that benefit the environment.
D. Social	Accessible and understandable by the user.

E. Ethical	Managing tree information and location.

Table 3.16 displays the different Health and Safety, Economic, Environment, Social, and Ethical for the project consideration. The considerations are important to the research because even if the system is completely functioning there might be uncertain things that might happen to the system.

Table 3.17. Design Process in Relation to Design Consideration Table

Design Process	Description	Design Considerations				
		A	B	C	D	E
Define the Problem	State the problem of the study					
Generate Related Information	Collect all data and information					
Analyze problem and Solution	Analyze the problem and decide on the best solution					
Requirements	A necessity for the design and important characteristic of the project					
Test and Implement the Solution	Test of the project and consideration for the restructuring of the project					

Table 3.17 shows the design process of the research. Each corresponding design process has its purpose to be able to accomplish the research. The X mark represents what is needed to be accomplished.

3.8 Functional and Test Cases

3.8.1 Test Process

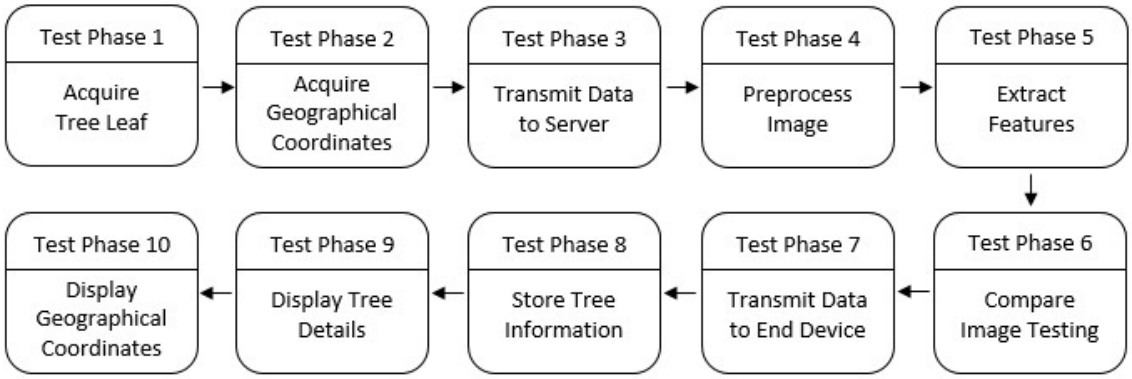


Figure 3.14. Function test phase

Figure 3.14 shows the overall test phase for each function. The test for each function would start with the acquire leaf tree function and would be followed by the acquire geographic coordinates function. The next function that would be tested is the transmit data to the server function going to the compare image function. Finally, the testing of the output wherein the tree details and the geographic coordinates are the final test phase.

Table 3.22. Acquire Tree Leaf & Geographical Coordinates Table

Test Case	Phase 1	Phase 2	Phase 3
1			
2			

Table 3.22 shows test phases and test cases of Acquiring tree leaf, Geographical Coordinates, and transmit data. In Test Phase 1, the image must be complete to be able to move to Phase 2, Test Phase 2 should be able to detect geographical coordinates to be able to complete the process and after completing Phase 1 and 2, Phase 3 will transfer the file to the server.

Table 3.23. Acquiring Tree Leaf Table (Test Phase 1)

Test Case	Place Condition	Leaf Condition	Che klist		Remarks
			Complete	Incomplete	
1					
2					
3					
4					
5					
6					

Table 3.23 shows Test Phase 1 as the different test cases when acquiring a tree leaf image. The leaves will be placed under bright and dark areas and the leaf would be in different conditions. Each place condition and leaf condition would be tested a total of

ten different times with the 5 different species and would be given the remark of either identified, unidentified or misinterpreted.

Table 3.24. Acquiring Geographical Coordinates Table (Test Phase 2)

Test Case	Location	Longitude	Latitude	Checklist		Remarks
				Complete	Incomplete	
1						
2						
3						
4						
5						
6						
7						

Table 3.24. Acquiring Geographical Coordinates Table (Test Phase 2) (Continuation)

Test Case	Location	Longitude	Latitude	Checklist		Remarks
				Complete	Incomplete	
8						
9						
10						

Table 3.24 shows Test Phase 2 as the different test cases when acquiring the geographical coordinates. The app should be able to acquire the longitude and latitude of the given location by the user.

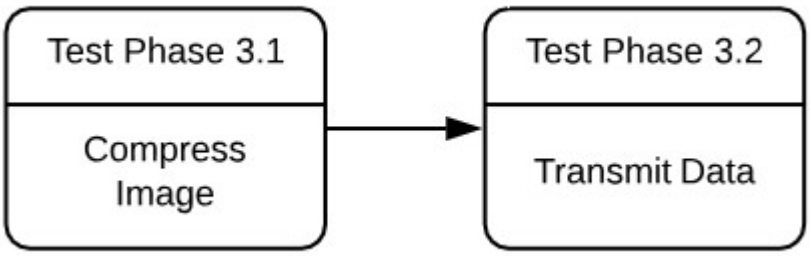


Figure 3.15. Transmit data to a server

Figure 3.15 shows Test Phase 3 that is divided into two phases, the compress image phase and the transmit data phase. The compressed image would be the part where the image would be compressed to different sizes of data and the transmit data would be whether or not the data would be received by the server. Both complete the entire function of the Transmit data function.

Table 3.25. Compress Image Data to Server Table (Test Phase 3)

Test Case	Image Data Size	Compress Data		Transmit Data		Remarks
		Yes	No	Complete	Incomplete	
1						
2						
3						
4						
5						
6						
7						

Table 3.25. Compress Image Data to Server Table (Test Phase 3) (Continuation)

Test Case	Image Data Size	Compress Data		Transmit Data		Remarks
		Yes	No	Complete	Incomplete	
8						
9						
10						

Table 3.25 shows Test Phase 3 as the different test cases when transmitting data to the server. The different size of data shows the compression of data and when the data is compressed, it should be able to transmit the data to the server.

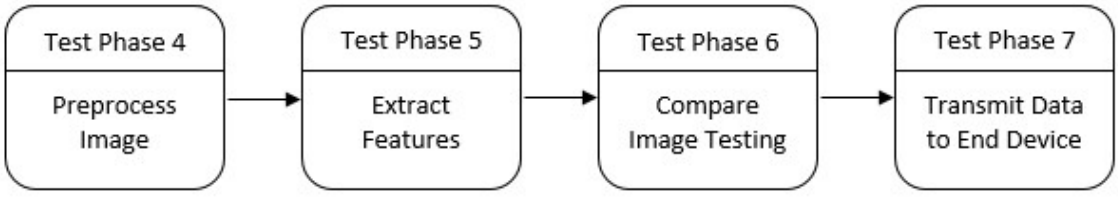


Figure 3.16. Backend process with the output result

Figure 3.16 shows Test Phases 4-7 for the testing for the backend section of the system. It follows the preprocessing of the image, then proceeds by feature extraction to be compared, and finally transmits the data back to the Android device by using a client-server connection. The test phases are rolled into one and the expected output must produce a classified tree species as a result of data requested from end devices. **Table 3.26.** Backend Process with Output Result

Test Case	Phase 4	Phase 5	Phase 6	Phase 7
1				
2				

Table 3.26. Backend Process with Output Result (Continuation)

Test Case	Phase 4	Phase 5	Phase 6	Phase 7
2				

Table 3.26 shows the continuation of Test Phases 1, 2, and 3. Test Phase 4 is the pre-processed image, which is the grayscale, removes image background, detects edges, and resize an image. Test Phase 5 is the extract image feature. Test Phase 6 is the compare image, and Test Phase 7 is the compressed data. Each test case in the test phase must be completed to be able to move to the next phases.

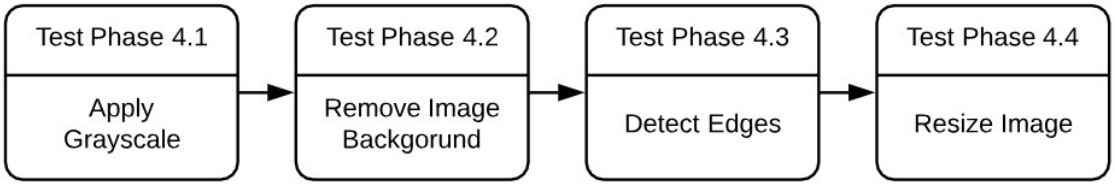


Figure 3.17. Pre-process image

Figure 3.17 shows the different phases within Test Phase 4. These include Grayscale, Removal of the image background, Edge Detection, and Resize of image. The following table below would show the different test cases for each phase.

Table 3.27. RGB Image to Grayscale (Test Phase 4.1)

Trial	Checklist		Remarks
	Able to detect tree leaf image	Unable to detect tree leaf image	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Table 3.27 shows Test Phase 4.1 as the RGB or the 3 common colors to detect grayscale. There is a specific number for each RGB color to be able to test if the grayscale is completed.

Table 3.28. Remove Image Background (Test Phase 4.2)

Trial	Background Setting		Checklist		Remarks
	Bright	Dark	Complete	Incomplete	
1					
2					
3					
4					
5					

6					
7					
8					
9					
10					

Table 3.28 shows Test Phase 4.2 as the test to be able to detect if the removed image background is completed. Setting up the bright and dark code to remove the image background.

Table 3.29. Detect Image Edges (Test Phase 4.3)

Trial	Set up Edges Code	Checklist		Remarks
		Complete	Incomplete	
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Table 3.29 shows Test Phase 4.3 to be able to detect if the image edges are detected successfully. This testing sets up the edge detection code to detect image edges.

Table 3.30. Resize Image (Test Phase 4.4)

Trial	Image Resized	Checklist		Remarks
		Complete	Incomplete	
1				
2				
3				
4				
5				

6				
7				
8				
9				
10				

Table 3.30 shows Test Phase 4.4 as the test to be able to detect if the image is successfully resized. This testing sets up the resizing image code to initiate image resize.

Table 3.31. Extract Image Features (Test Phase 5)

Trial	Set up Extract Image Feature Code	Checklist		Remarks
		Complete	Incomplete	
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Table 3.31 shows Test Phase 5 to be able to detect if the image featured is successfully extracted. This testing sets up the extract image feature code to extract features key points from the image.

Table 3.32. Compare Image with Predefined Tree Features (Test Phase 6)

Trial	Features from an acquired image	Pre-defined feature from a tree leaf	Checklist			Remarks
			Identified	Unidentified	Misinterpreted	
1	Acacia					
2						
3	Banaba					
4						
5	Molave					
6						

7	Mahogany					
8						
9	Yemane					
10						

Table 3.32 shows Test Phase 6 as the test if the feature extract image is identified, unidentified, or misinterpreted to the pre-defined feature.

Table 3.33. Transmit Data to Android Device Table (Test Phase 7)

Test Case	Image Data Size	Compress Data		Checklist		Remarks
		Yes	No	Complete	Incomplete	
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Table 3.33 shows Test Phase 7 as the different test cases when transmitting data to the mobile devices. The different sizes of data show the compression of data and when the data is compressed it should be able to transmit the data to the end devices of Android smartphones.

3.9 System Test

3.9.1 Test Process

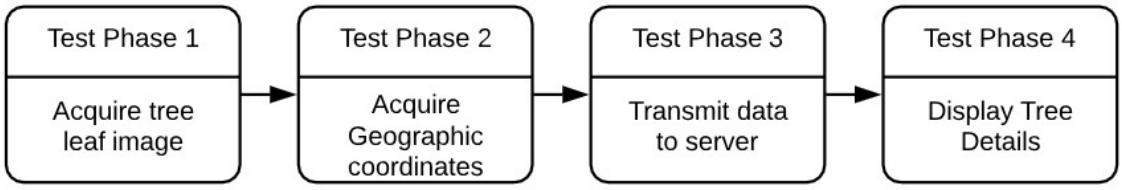


Figure 3.18. System test phase

Figure 3.18 shows the overall system test phase. It shows the flow to test the overall accuracy, user-friendliness, consistency, and responsiveness of the system. Once the flow is followed, the output of the system would determine if the objective has been achieved.

Table 3.34. Accuracy Table Testing

No.	Accuracy test	Checklist		Remark
		Yes	No	
1	Able to identify tree leaf (Acacia) minimum of 10 samples			
2	Able to identify tree leaf (Banaba) minimum of 10 samples			
3	Able to identify tree leaf (Molave) minimum of 10 samples			
4	Able to identify tree leaf (Mahogane) minimum of 10 samples			
5	Able to identify tree leaf (Yemane) minimum of 10 samples			

Table 3.34 shows the system testing to meet the objective for accuracy. The proponents must test multiple times whether the acquired image was identified accurately and successfully with their respective species.

Table 3.35. User-friendly Table Testing

No.	User-friendly test	Checklist		Remark
		Yes	No	
1	Navigation is easily Identifiable			
2	Navigation labels are clear			
3	Navigation labels are concise			
4	Information is clear, enough, and uncluttered			
5	Icons are properly and represented			

Table 3.35 shows the system testing to meet the objective for user-friendliness. The testing focuses more on the functionality of the UI if it is working as it is designed and provides clarity and distinctiveness for users to easily navigate throughout the application. **Table 3.36.** Consistency Table Testing

No.	Consistency test	Checklist		Remark
		Yes	No	

1	Able to identify tree leaf (Acacia) with varying distance minimum of 30 samples			
2	Able to identify tree leaf (Banaba) with varying distance minimum of 30 samples			
3	Able to identify tree leaf (Mahogane) with varying distance minimum of 30 samples			
4	Able to identify tree leaf (Molave) with varying distance minimum of 30 samples			
5	Able to identify tree leaf (Yemane) with varying distance minimum of 30 samples			

Table 3.36 shows the system testing to meet the objective for consistency. The proponents must test whether it can recognize the tree of the leaf at a longer distance.

The farther the distance or screen ratio to be captured on the image the better. **Table 3.37.** Responsive Table Testing

No.	Responsive test	Checklist		Remark
		Yes	No	
1	Able to send data using packet WIFI within 1 to 5s			
2	Able to send data using 3G within 1 to 5s			
3	Able to send data using 4G within 1 to 5s			
4	Able to send data using LAN within 1 to 5s			

Table 3.37 shows the system testing to meet the objective for responsiveness. The testing focuses more on the latency or process time for a request to successfully respond throughout the system. The proponents will measure the time it takes for a request to finish with different wired and wireless connectivity technologies. The lower the latency the better.

3.10 Evaluation Procedure and Criteria for Overall System

$$Individual\ Criteria\ Rating\ (\%) = \left(\frac{Positive\ Criteria\ Rating}{Total\ Number\ of\ Specific\ Critera}\right)\ x\ 100\% \quad \textbf{Eq. 3.12}$$

Equation 3.12 aims to calculate the correctness of the system in giving information. For the accuracy of the system, Equation 3.12 determines the performance rating of each criterion to identify which of the criteria would need improvement over the other. The formula requires the positive criteria rating of each criterion per system test

and divide it by the total number of rating for that specific criterion. The result would show if the function was able to attain a high rating on that specific criterion.

$$Overall\ Criteria\ Rating\ (\%) = \left(\frac{Total\ Individual\ Critera\ Rating}{Total\ Number\ of\ Criteria}\right)\ x\ 100\%$$

Equation 3.13 represents the success rate where it would show the rating of the specific objective. The formula consists of the total individual rating divided by the total number of criteria which would result in the average of a system function. This would show if a specific system function can attain a high rating. The objectives would be divided into 4 ratings depending on the result, ranging from excellent, very good, fair, and poor.

Table 3.38. Evaluation of the Objectives of the Study for Accuracy

Objective	#Trials	Number of Success rate Table 3.34	Success rate	System Rating
Ability to identify different tree leaves		*Criterion 1*	*Criterion 1*	
		Criterion 2	*Criterion 2*	
		Criterion 3	*Criterion 3*	
		Criterion 4	*Criterion 4*	
		Criterion 5	*Criterion 5*	

Table 3.38 shows the evaluation for the accuracy of the system. As shown in Equation 3.12 each criterion would be evaluated to determine if the accuracy of the system has a high rating. The total of each criterion would then be averaged using Equation 3.13.

Table 3.39. Evaluation of the Objectives of the Study for Reliability

Objective	#Trials	Number of Success rate Table 3.34	Success rate	System Rating
Ability to the system to identify correct tree species		*Criterion 1*	*Criterion 1*	
		Criterion 2	*Criterion 2*	
		Criterion 3	*Criterion 3*	
		Criterion 4	*Criterion 4*	
		Criterion 5	*Criterion 5*	

Table 3.39 shows the evaluation for the reliability of the system. Similar to the evaluation of Table 3.38, it would also apply both Equation 3.12 and Equation 3.13 in determining the rating for acquiring the system's reliability incorrect identification of tree species.

Table 3.40. Evaluation of the Objectives of the Study for Consistency

Objective	#Trials	Number of Success rate Table 3.36	Success rate	System Rating
Ability to identify tree leaves at varying distances		*Criterion 1*	*Criterion 1*	
		Criterion 2	*Criterion 2*	
		Criterion 3	*Criterion 3*	
		Criterion 4	*Criterion 4*	
		Criterion 5	*Criterion 5*	

Table 3.40 shows the evaluation for the consistency of the system. It would still apply Equation 3.12 and Equation 3.13 in determining if the function of the system has a high rating. The criterion is based on Table 3.36.

Table 3.41. Evaluation of the Objectives of the Study for Responsiveness

Objective	#Trials	Number of Success rate Table 3.37	Success rating	System Rating
Responsiveness of data transmission		*Criterion 1*	*Criterion 1*	
		Criterion 2	*Criterion 2*	
		Criterion 3	*Criterion 3*	
		Criterion 4	*Criterion 4*	
		Criterion 5	*Criterion 5*	

Table 3.41 shows the evaluation for the responsiveness of the system. The evaluation of the Responsiveness would be based on Table 3.37 and would also be rated using Equation 3.12 and Equation 3.13, this would show if the system is responsive in terms of the set criteria.

Table 3.42. Evaluation of the Objectives of the Study for User-friendliness

Objective	#Respondents	Number of Success rate Table 3.35	Positive Rating	System Rating
User-friendliness of the Android application		*Criterion 1*	*Criterion 1*	
		Criterion 2	*Criterion 2*	
		Criterion 3	*Criterion 3*	
		Criterion 4	*Criterion 4*	
		Criterion 5	*Criterion 5*	

$$Individual\ Criteria\ Response\ (\%) = \left(\frac{Positive\ Response}{Total\ No.\ of\ Specific\ Criteria}\right) x\ 100\% \quad \textbf{Eq. 3.14}$$

Equation 3.14 shows the individual criteria response when evaluating for user-friendliness. Each response would be divided into the total number of specific criteria to find the area that needs improvement in terms of user-friendliness.

$$Overall\ Criteria\ Response\ (\%) = \left(\frac{Total\ Individual\ Response}{Total\ Number\ of\ Criteria}\right) x\ 100\%$$

Equation 3.15 represents the total response concerning the application. Each criteria response is tallied and is then divided into the total number of criteria to find the average response to the application. We would then be able to identify if the application is deemed user-friendly by the users.